Dealer Spreads in the Corporate Bond Market: Agent vs. Market-Making Roles

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Abstract

We use the large proportion of "riskless-principal-trades" to investigate corporate bond dealer-spreads arising from dealers' roles as agents or market-makers. We find that agent-related-spreads are comparable in magnitude to market-making-spreads, and consistent with liquidity-search and customer-interface-costs/benefits; while market-making-spreads are consistent with inventory/asymmetric-information costs. In particular, agent-related spreads are lower for likely automated electronically-executed trades with clearly lower search costs. Dealers also appear to benefit informationally through interfacing directly with customers, and post tighter spreads in non-automated settings. Our results show that earlier TRACE studies have typically underestimated average trading costs by ignoring the separate agent and market-making roles of dealers.

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Dealer Spreads in the Corporate Bond Market: Agent vs. Market-Making Roles

Financial intermediaries in the U.S. corporate bond market, as in most over-the-counter markets, operate as 'broker-dealers' with two functions. As brokers, they serve as agents for individual or institutional traders (hereafter "customers"), providing market access and counterparty search services. As market-makers, they stand ready to buy (or sell) on their own account as principals when a customer wants to sell (or buy). In a particular trade, a dealer may function purely as an agent, purely as a market-maker, or in both capacities. We hereafter use the terms "agent-dealer", "principal-dealer", and "dual-capacity-dealer" respectively for a bond dealer functioning in a specific trade purely as an agent, purely as a market-maker, or in a dual capacity as both agent and market maker.[1] If a trade goes through an agent-dealer, the customer's total transaction costs include two spreads: that of the agent-dealer on a "riskless principal trade" plus that of the principal-dealer. While agent-dealer spreads arguably impound counterparty search costs (Duffie, 2012) and customer interface costs and benefits, the principal-dealer's spread impounds market-making costs, i.e., inventory and asymmetric information costs and risks. Spreads of dual-capacity dealers impound both market-making costs and customer interface costs and benefits, but not counterparty search costs.

In most over-the-counter market data, it is difficult to distinguish when dealers are functioning as agent-dealers, principal-dealers, or dual-capacity dealers; and empirical studies of dealer spreads have generally treated dealer spreads as primarily representing market making costs.[2] However, in the corporate bond market, TRACE data enable distinguishing trades in which a dealer functions purely as agent, purely as market-maker, or in dual-capacity. In this paper, we use this

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[1] The agent and principal roles may differ from bond to bond and from trade to trade, i.e., a dealer may be a principal-dealer for one set of bonds or trades, an agent-dealer for another set of bonds or trades, and a dual-capacity-dealer for a third set of bonds or trades.

[2] See, e.g., Bessembinder (1994) and Lyons (1995) for FX markets; Huang and Stoll (1996), Barclay, Christie, Harris and Schultz (1999), and Huang (2002) for NASDAQ; Hansch, Naik and Viswanathan (1998, 1999) and Naik and Yadav (2003a) for London Stock Exchange; Naik and Yadav (2003b) for London government bond market; and Harris and Piwowar (2006) for the US municipal bond market. There is little empirical evidence on the dealer's role as agent: the results of Ashcraft and Duffie (2007) for bilateral trades between banks in the Federal Funds market are consistent with theoretical search-based approaches, but do not enable inferences on intermediary transaction costs.

opportunity to separately investigate dealer spreads arising from their market-making and agent roles, analyze the determinants of each, and explore associated hypotheses and implications.

We are able to separately estimate the agent-related and market-making components of dealer spreads by utilizing the sizeable subset of U.S. corporate bond trades in which some dealers act purely as agents which are an estimated 39.3% of customer buys and 34.2% of customer sells in our two-year sample period. Other studies (e.g., Harris, 2015) find similar proportions. If dealer A receives a customer buy order for a bond that it does not have in inventory, it must obtain the bonds from another dealer B who does. In this situation, dealer A often engages in a "riskless principal trade" in which it arranges to: 1) buy the bonds from dealer B and 2) *simultaneously* resell the bonds to the customer. Likewise, if dealer C receives a sell order for a bond that it does not wish to keep in inventory, C may execute a riskless principal trade in which it buys the bonds from the customer and *simultaneously* resells to dealer B. In these riskless principal trades, dealers A and C do not carry any price or inventory risk, and serve purely as agents of their customers. Hence, their spread represents compensation for their agent-dealer services of searching for the counter-party with the best price and managing customer orders and relationships. On the other hand, in these cases, Dealer B serves purely as a market maker, bearing all inventory and asymmetric information costs and risks, but no search or customer interface costs. Thus, total customer trading costs on these trades consist of two spreads: 1) that of the customer-interfacing agent-dealer, A or C, who arranges the riskless principal trade and acts purely as an agent, and 2) that of the liquidity providing principal-dealer, B, who holds bonds in inventory and acts purely as a market-maker.

On the other hand, the majority of bond trades – 60.7% (65.8%) of customer buys (sells) in our sample period – do not go through a separate agent-dealer. In these trades, a dealer sells directly to a customer from its bond inventory or retains bonds bought directly from a customer in its inventory. In these cases, the dealer acts in dual capacity – as an agent for the customer and as a market making principal and the customer bears the search costs. In this paper, we estimate agent-dealer, principal-dealer, and dual-capacity-dealer spreads, and analyze how they vary with the factors that are (or are not) relevant to the specific economic function (i.e., agent, or market-maker, or both) being fulfilled by them. Since the three roles are quite different, we expect the determinants of their respective spreads to differ.

The market-making component of dealer costs represents dealer compensation for inventory costs and the asymmetric information risk of trading with more informed traders.[3] Since principal-dealers and dual-capacity-dealers make markets and agent-dealers do not, we expect principal-dealer and dual-capacity-dealer spreads, but not agent-dealer spreads, to be positively related to measures of market price risks -- specifically: the VIX index of expected equity market volatility and the MOVE index of expected interest rate volatility. Anticipating that price risks, and therefore inventory and asymmetric information costs, are higher on lower rated bonds and longer duration bonds, we also expect principal-dealer and dual-capacity-dealer spreads to be higher on lower rated and longer maturity bonds.

Since the agent-related component of dealer costs represents compensation for counterparty search costs and customer interface costs – rather than a premium for risk – we expect little correlation between agent-dealer spreads and market price risk variables such as the VIX and MOVE indices. On the basis of theoretical search cost models, we expect search costs, and hence agent-dealer spreads, to be higher for assets with low liquidity -- specifically bonds with low trading volume, low ratings, and longer terms-to-maturity.[4] Additionally, and importantly in the context of the significant changes currently underway in fixed income trading microstructure, we expect agent-dealer search costs, and hence the associated agent-dealer spreads, to be lower for trades where dealers and/or their customers utilize electronic systems in which trade execution is partially or fully automated – like electronic order-matching systems with firm standing quotes, or electronic request-for-quote systems with automated-response mechanisms – as against trades needing to query multiple dealers for quotes prior to the trade through voice-based methods, or electronic request-for-quote systems *without* automated-response mechanisms.[5] We hereafter label such trades as "automated electronically-executed trades".

While both principal-dealers and dual-capacity-dealers have market-making costs, i.e., inventory and asymmetric information costs and risks, dual-capacity-dealers also have customer-interface costs and benefits. Thus, the difference between the two on similar bonds and trades

[3] See, e.g., Ho and Stoll (1983); Glosten and Milgrom (1985); Glosten and Harris (1988); and Stoll (1989).

[4] Theoretical models of search costs are relatively recent - pioneered in the finance literature by Duffie, Garleanu, and Pedersen (2005), and include Weill (2007), Vayanos and Weill (2008), Lagos and Rocheteau (2009). Rocheteau and Weill (2011), Afonso (2011), and Lagos, Rocheteau and Weill (2011).

[5] The July 2016 Tabb Group online publication titled "Best Execution in Fixed Income: A Work in Progress" (https://www.theice.com/publicdocs/knowledgecenter/BestExecutioninFixedIncome.pdf) written by Anthony Perotta Jr. and Colby Jenkins provides on pages 3-4 an excellent discussion of the electronic fixed income trade execution systems that have proliferated after the financial crisis in the wake of proprietary trading restrictions.

provides a measure of *net* customer interface costs of dual-capacity dealers. While these net costs include explicit order-processing and other costs of managing customer relationships, they likely also impound customer interface related benefits. In particular, by interacting directly with the trader, the dual-capacity dealer may potentially be able to know the identity of the trader, judge the extent to which the trader is informed, and (at least) partially infer the private information impounded in the trade, and adjust spread accordingly.[6] On the other hand, in trades routed through an agent-dealer, the associated principal-dealer does likely not get access to this information.[7] We expect the dual-capacity-dealer's customer interface benefit to be greater (and net customer interface costs to be lower) for bonds with high information asymmetry, e.g., lower rated bonds or bonds that are rarely traded, and also when market uncertainty as represented by the VIX and MOVE indices is high.

We also examine how explicit trading costs differ depending on whether a trade involves both agent- and principal-dealers, or is conducted directly with a dual-capacity-dealer. Non-dealer customers potentially bear unobservable internal search costs on trades with dual-capacity-dealers.

Most earlier TRACE data based estimates of bond trading costs treat the customer trade with the agent-dealer, and the agent-dealer trade with the principal-dealer, as independent and separate trades; and not linked inter-connected operations involved in supplying liquidity to the same customer trade. Thus, they underestimate total customer trading costs. Additionally, hypotheses tests treat all customer trades with dealers as trades with principal-dealers. Extending the model of Edwards, Harris, and Piwowar (2007) (hereafter "EHP (2007)"), we distinguish between agent-dealer, principal-dealer, and dual-capacity dealer spreads, and explore how recognizing that some trades involve two spreads impacts the resulting transaction cost estimates.

We provide several important results. Our first major contribution is to estimate and test hypotheses on agent-dealer spreads. One, agent-dealer spreads are sizable and comparable in magnitude to principal-dealer spreads, implying that dealers' agent-related costs, i.e., search and customer interface costs, are comparable to dealers' market-making costs. Two, as hypothesized, since they do not bear any inventory and asymmetric information risks, agent-dealer spreads are not

[6] The potential ability of dealers to infer private information from customer orders and trades is modeled, for example, in Naik, Neuberger, and Vishwanathan (1999).

[7] The agent-dealer does not directly derive any benefits from trying to distinguish between informed and uninformed traders since she does not maintain an inventory in this bond, and thus faces no asymmetric information risk.

related with measures of market uncertainty – specifically the VIX index and the MOVE index of interest rate volatility. Three, agent-dealer spreads are lower for trades that provide a direct proxy for lower search and order processing costs: i.e., trades in which dealers and/or their customers utilize electronic systems in which trade execution is likely partially or fully automated. We also document other supporting evidence on the relationship of agent-dealer spreads with search and order-processing costs through proxies for liquidity: agent-dealer spreads increase as bond specific trading volume decreases, and for lower rated and longer maturity bonds. Finally, while principal-dealer and dual-capacity-dealer spreads decline sharply as trade size increases (consistent evidence from earlier studies and the presence of fixed costs), agent-dealer spreads are roughly constant (or even increase slightly) until trade size reaches a threshold, suggesting that agent-dealers devote less search effort to smaller trades and/or that larger traders have greater bargaining power.

Our second contribution is to cleanly test hypotheses on market-making. As expected from what we know from extant literature, since they bear inventory and asymmetric information risks, both principal-dealer spreads and dual-capacity dealer spreads are significantly positively related with the VIX and the MOVE indices of volatility and market uncertainty. They are also negatively related to bond specific trading volume – a proxy for liquidity – and are higher for lower rated and longer maturity bonds for both liquidity and risk reasons. However, importantly, we find that principal-dealer spreads for automated electronically-executed trades are slightly higher, suggesting that principal-dealers may not post their tightest quotes in systems where standing quotes can be automatically executed. This is not surprising since the options that principal dealers provide to the market through their free-standing or pre-programmed quotes are long-horizon options – in contrast to the short shelf-life options they provide when they revert with a quote in response to a specific request-for-quote in a non-automated framework.

Our third major contribution is to investigate the dual-capacity feature of dual-capacity-dealers. We find that dual-capacity-dealers extract significant information-related benefits from their direct interaction with trading customers. Our evidence indicates that this benefit is greatest on large trades, when market uncertainty is high, and on issues with more information asymmetry. Importantly, we also find that, except for very small trades, customers face significantly lower total explicit trading costs if they trade directly with a dual-capacity-dealer rather than through an agent-dealer. In addition, as expected, given that their ambit includes market-making, we find that, similar to principal-dealers, dual-capacity-dealer spreads are related positively with VIX and MOVE volatility indices, negatively related to bond specific trading volume, and are higher for lower rated and longer maturity bonds.

Finally, we find that most earlier estimations of bond market transaction costs from TRACE data tend to significantly underestimate total transaction costs because they failed to recognize that customer transactions often involve two component trades and spreads. This helps explain the wide variation in previous bond transaction cost estimates.

Our results have important implications for the extensive extant empirical research on dealer spreads in other OTC markets since these studies have also not separated dealers' agent-related and market-making roles and costs and have tended to focus solely on dealers' making-making role. The results in this literature need to be re-interpreted in the context of the large agent-related costs that we find, costs that meaningfully vary with several relevant economic factors in a manner that is similar to market-making costs in some cases, and differently in many other cases. Second, our results on the possible informational benefits to dual-capacity-dealers of directly interacting with customers – should also be applicable to other OTC markets. Third, with traditional OTC voice-based request-for-quote markets being increasingly substituted by automated electronic execution methods, our result that principal-dealers may not post their best quotes in systems where standing quotes can be automatically executed, may again have wider applicability to other OTC markets.

The remainder of this paper is organized as follow. Section 1 briefly reviews the extant empirical literature on bond market transaction costs. Section 2 describes the TRACE data. In section 3, we develop our hypotheses regarding likely determinants of agent-dealer, principal-dealer and dual-capacity-dealer spreads. In section 4, we estimate and analyze agent-dealer spreads on riskless principal trades, and principal-dealer spreads on paired agent-dealer trades. In section 5, we estimate and analyze dual-capacity-dealer spreads on trades directly between dual-capacity-dealers and non-dealer customers. In section 6, we use all trades and our extension of the EHP (2007) model, to estimate and analyze all three spreads. We also explore the costs and benefits to dual-capacity-dealers of the direct customer interface. In section 7, we explore the cost differences between transactions involving both agent- and principal-dealers and those directly with dual-capacity dealers. Section 8 explores the bias in extant transaction cost estimates from failing to recognize that some trades involve two linked transactions and spreads, and attempts to explain the great disparity among previous bond transaction cost estimates. Finally, Section 9 concludes.

1. Relevant Literature on Corporate Bond Transaction Costs

In recent years, most empirical estimates of bond transaction costs have been based on either of two databases: (1) insurance company trades from the National Association of Insurance Commissioners (NAIC) and (2) the National Association of Securities Dealer's Trade Reporting and Compliance Engine (TRACE), which since 2005 reports virtually all bond trades by security dealers.[8] As discussed in section 8, transaction costs estimates based on the insurance company trades data tend to be much lower than those based on TRACE.[9] While our results are based on TRACE, they help explain this difference. Most studies also employ either of two estimation approaches: (1) estimating transaction costs as the difference between paired purchase and sale prices (which normally differ by trade time and size) of the same bond and (2) multivariate regression in which each transaction price is regressed on variables to control for buys versus sells, bond characteristics, and market conditions.[10] We employ the first procedure in sections 4 and 5 below and the second in the remaining sections.

The papers most relevant to our paper are: Harris (2015), Choi and Huh (2017), Ruzza (2017), Schultz (2017), Bao et al. (2016) and Bessembinder et al. (2017).[11] By examining paired dealer-customer buy and sell trades, Choi and Huh (2017) explore the liquidity provided by customers. Ruzza (2017) focuses on informativeness and adverse selection finding that the prices (relative to average TRACE trade prices for that bond that day) received by non-bank customers on paired trades (involving, in our terminology, both an agent-dealer and a principal-dealer), are worse

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[8] Hong and Warga (2000) report that spreads on the NYSE's electronic bond market are comparable to those from the NAIC database. According to Bessembinder, Maxwell, and Venkataraman (2006) the insurance database accounts for about 12.5% of the dollar volume on TRACE. Since NAIC's average trade size is much larger than TRACE's, the NAIC share of trades is considerably smaller.

[9] Henderschott et al. (2015) and O'Hara et al. (2016) explore execution costs on insurance companies' bond trades. Henderschott et al. (2015) find that execution costs are greater the smaller the insurance company, the smaller the bond dealer, and the smaller the network. O'Hara et al. (2016) find that, holding bond and other factors constant, insurance companies that trade a lot receive better prices than those that trade less often. Interestingly for our paper, O'Hara et al. (2016) also report that the OTC bond market is highly concentrated with the top three dealers in a particular bond accounting for an average of 92% of insurance company trading volume in that bond.

[10] Examples of the former include Hong and Warga (2000), Chakravarty and Sarkar (2003), Goldstein Hotchkiss, and Sirri (2007), Feldhutter (2012), O'Hara, Wang, and Zhou (2016) and Henderschott, Li, Livdan, and Schurhoff (2105). Examples of the latter include EHP (2007), Goldstein Hotchkiss, and Sirri (2007), Bessembinder, Maxwell, and Venkataraman (2006), Harris (2015), Choi and Huh (2017),and Ruzza (2017).

[11] Also related to our paper, Zitzewitz (2011) finds that trading costs on his "paired trades" are higher than those on unpaired and are split roughly equally between the paired and ultimate dealers. Unlike the other papers, Sirri (2014) examines trading costs on municipal, not corporate, bonds. He recognizes that some trades involve two dealers and two spreads but, by allowing up to 30 days for the first dealer to pass the bonds on to another dealer, both dealers in his paired trades bear some inventory risks, and so differ from our pure agent-dealer and pure principal-dealer classifications.

than those received in (unpaired) trades with a dual-capacity dealer. He interprets this as evidence of dealers shifting the risk of informed trading to their clients. Based on a sample of 70 very actively traded bonds, Schultz (2017) finds that sellers receive higher prices and buyers pay lower prices on riskless principal trades. Bessembinder et al. (2017) examine spreads on what we term dual-capacity dealer trades by the more active dealers. They find no evidence that transaction costs on trades with dual-capacity dealers increased after the financial crisis and imposition of the Volcker rule but that measures of liquidity and trade quality declined. Bao et al. (2016) compare how spreads and liquidity measures change around rating downgrades before and after the Volcker rule, concluding that liquidity declined after the Volcker rule. They also find that for banker-dealers affected by the Volcker rule, the proportion of trades handled on an agency basis following downgrades increased after the crisis and imposition of the Volcker rule.

In an important paper, Harris (2015) compares TRACE trade prices with the best bid and ask quotes at the same time obtained from Interactive Brokers (IB). Though Harris (2015) acknowledges that many of the IB quotes may be indicative, he argues that most are firm. His IB data enable calculation of transaction cost measures that are qualitatively different from those in the literature thus far, including, in particular, effective spreads. Focusing on best execution, he finds that approximately 43% of the trade prices on TRACE are "trade-throughs", i.e., buys above the best ask or sales below the best bid, indicating that the customer did not receive the best price in these cases. Several of our findings below are consistent with his argument that agent-dealers do not always find the best possible price for their customers.

While several of these papers touch on aspects of our paper, their focuses are different from ours. Although some estimate dealer spreads for dealers in general, none separately investigate the dealer spreads arising from their market-making and agent roles, nor explore associated hypotheses regarding their respective determinants. Excepting Schultz (2017) whose data is limited to 70 very actively traded bonds, none have compared trading cost for bonds that go through an agent-dealer with those directly with a dual-capacity dealer. None have explored the possible advantage to dual-capacity dealers of knowing the customer's identity.

There have been significant changes in the trading microstructure of fixed income markets over the past decade which have attracted research. One is the withdrawal of bank-affiliated market makers from the fixed income markets documented and explored by Bessembinder et al. (2017), Bao et al. (2016), Choi and Hui (2017) and others. A second is the proliferation of different types of electronic systems in addition to traditional voice-based trading: e.g., Click-to-Trade, All-to-All, Request-for-quote), and Request-for-Stream systems.[12] On one hand, dealers/customers can utilize electronic execution systems in which trade execution is partially or fully automated, like electronic order-matching systems with firm standing quotes, or electronic request-for-quote systems with automated-response mechanisms. On the other hand, a dealer/trader can trade by querying multiple dealers for quotes prior to the trade through voice-based methods, or electronic request-for-quote systems *without* automated-response mechanisms. We distinguish between these two types in our empirical analyses, labeling the former as "automated electronically-executed trades".

2. Data

Our analysis is based on corporate bond trades reported on the Trade Reporting And Compliance Engine ("TRACE") over the period November 3, 2008 to December 31, 2010. On November 3, 2008, TRACE started attaching "S", "B", and "D" codes to reported trades; with sales of bonds by a dealer to a (non-dealer) customer given an "S" designation, purchases by a dealer from a customer designated "B", and interdealer trades designated "D". Our data time period includes the financial crisis and its immediate aftermath but not imposition of the Volcker rule.

Bond characteristics, such as coupon, maturity, and ratings, were obtained from Mergent's FISD database. For inclusion in our sample, we require that the bond be a non-convertible, non-putable, industrial bond or note denominated in US dollars with fixed (possibly zero) coupon, $1000 par value, semi-annual coupon payments, and at least three years to maturity as of November 1, 2008 which is neither in default nor has a tender offer outstanding. 3859 bonds meet these requirements. Bonds are dropped from the sample when they default, are called or retired, or maturity drops below three years. Expanding on Dick-Nielsen (2009) and Dick-Nielsen (2014), we drop TRACE trade observations if: 1) the trade is later corrected or canceled, 2) settlement is over a week in the future, 3) it is a "when issued" or "special price" trade, 4) there is an unreported

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[12] Specific Examples of such systems include Liquidnet, MTS Bonds Pro, TMC Bonds, Tradeweb or Tradeweb Direct, Bloomberg, KCG Bondpoint, and MarketAxcess. Most descriptions and discussion of these electronic systems are available in industry articles rather than in academic publications or working papers. Besides the Tabb Group publication cited in footnote 7, useful references could be: (a) "The Trade" Issue 40, Jan-Apr 2014, Page 47-52, www.thetradenews.com; (b) "Digitization shakes up bond markets", The Economist, April 22, 2017; and (c) "There is a problem with electronic bond trading", Business Insider, March 17, 2017.

commission,[13] 5) a special sale condition is attached, 6) it is an "as of" trade, or 7) the price is less than $25 per $100 par value (which we regard as being in default even if there is no indication of default on FISD).[14]

TRACE's S trades include both dual-capacity-dealer sales (from the dealer's inventory direct to a customer) and agent-dealer riskless-principal sales of bonds (simultaneously bought in a separate trade from a principal-dealer). Likewise, TRACE's B trades include both dual-capacity-dealer purchases and agent-dealer riskless principal purchases. As in Zitzewitz (2011), Harris (2015), and Bessembinder et al. (2017), if a S trade and a D trade are within a minute of each other and are for exactly the same quantity, we assume that the S trade is an agent-dealer's riskless principal sale to the customer and that the D trade is the agent-dealer's accompanying purchase of the bonds from a principal-dealer. This also makes it possible to sign the paired D trade as a principal-dealer sale at it's ask price. Likewise, if a B and a D trade are for exactly the same quantity and within a few seconds of each other, we assume that the B trade is an agent-dealer riskless principal purchase, and the accompanying D trade is a principal-dealer purchase (and agent-dealer sale) at the principal-dealer's bid price. *In this context: price differences between paired S and D trades are the spreads of agent-dealers on customer buy orders, price differences between paired D and B trades are the spreads of agent-dealers on customer sell orders, and price differences between D trades paired with S trades and D trades paired with B trades represent round-trip principal-dealer spreads.*

Our non-regulatory version of TRACE does not contain dealer identifiers. Thus it is conceivable that two S and D (or B and D) trades within one minute of each other involve different dealers and are paired in error. However, this seems highly unlikely given the infrequency of corporate bond trades and our requirement that the two trade sizes be the same. On an average day,

[13] TRACE flags observations in which dealers report that there was a separate commission that is not included in the price but TRACE does not report what the commission was. We drop the few observations that indicate that there is a commission since we cannot observe the amount.

[14] As a final check, we compare yield-to-maturity (YTM) reported on TRACE with the YTM calculated from the trade price and settlement date reported on TRACE together with the coupon and maturity from the FISD database, and drop the observation if the two yields differ by more than ten basis points. For most observations, the two YTMs are virtually identical but for 1.8% of the sample, the two yields differ by more than ten basis points indicating that either: 1) the TRACE price or YTM is incorrect, 2) the bond CUSIP is incorrect, 3) the FISD coupon or maturity date is incorrect, or 4) there was a commission (which is included in the TRACE YTM but not the price) on the trade despite the fact that the TRACE data indicates no commission. It is apparent that in some cases either the TRACE CUSIPs or the FISD dates are incorrect since trades are reported before the bond was issued or after it was retired. If TRACE does not report a yield, we assume the price is correct.

more bonds do not trade than trade and for those that do trade, the the median number of additional trades that day is only three. Excluding cases when same size S and D or Band D trades are within one minute of each other, the average time between trades of the same bond is 11.9 hours and the percentage of times the two trades are the same size is 22.6% Hence, it seems highly unlikely that two unrelated trades of exactly the same size occur within one minute of each other by chance.

To illustrate our trade classification procedure, consider the sample of all trades in Alcoa's 5.5% 2017 notes between 9:24 AM and 12:01 PM on July 29, 2010 reproduced in Table 1. Trade characteristics reported by TRACE are shown in columns 2-6. Consider trades 1 and 2 in Table 1. An interdealer trade, D, at 9:24:55 for 10 bonds is followed one second later by a purchase, B, from a customer for 10 bonds. It seems apparent that in this case a dealer received a sell order from a customer for 10 bonds. Not wishing to keep the bonds in inventory, the dealer searched for another dealer to provide liquidity, and accordingly arranged a riskless principal trade in which he arranged to resell the bonds to the other dealer and then executed the two orders roughly simultaneously. We designate the agent-dealer's purchase from the customer (trade 2) as an "agent-dealer purchase" and the resale to the principal-dealer (trade 1) as a "principal-dealer purchase." In trade 3, the B trade for 25 bonds is not accompanied by a D trade, so is classified as a "dual-capacity-dealer purchase" directly from a non-dealer customer. Trades 4 and 5 are D and S trades respectively for exactly the same number of bonds, 16, at the same time, 10:23:10. Thus it appears that a dealer received a buy order from a customer for bonds it did not have, so arranged to purchase the bonds from a principal-dealer for $102.042 and to resell to the customer for $103.419. We classify the D trade (trade 4) as a "principal-dealer sale" and the S trade (trade 5) as an "agent-dealer sale."

Trade 6 is a standalone D trade for 3000 bonds. Since there is no accompanying B or S trade, we classify it as "interdealer-unpaired". In this case, unlike the D trades in trades 1 and 4, we cannot determine if trade 6 was at the bid or ask.[15] In trades 7, 8, and 9 we again have a B trade accompanied by a D trade but note that there are two D trades for the same number of bonds at the same time and price. Similarly, in trades 14, 15, and 16, an S trade is accompanied by two D trades for the same quantity at the same price. As explained by Sirri (2014), this trade apparently involved two agent-dealers in which dealer A bought the bonds and immediately sold to dealer B who immediately sold to dealer C who kept the bonds on its books. Since our focus is on total agent-

[15] These interdealer-unpaired trades represent risk-sharing trades of market-making dealers among themselves and do not directly impact dealer spreads for trades with customers, whether made directly or through an agent-dealer.

dealer spreads (whether involving one agent-dealer or several), we remove D trades 9 and 16 from our sample. There are 296,307 such duplicate or intermediate trades leaving us with a sample of 5,839,480 trades.[16]

To summarize, like Zitzewitz (2011), Harris (2015) and Bessembinder et al. (2017), our decision rule is: if a S (B) trade is accompanied within one minute by a D trade for exactly the same amount, we designate the S (B) trade as an agent-dealer sale (agent-dealer purchase) and the paired D trade as a principal-dealer sale (principal-dealer purchase). S and B trades not accompanied within one minute by a D trade of the same amount are designated as dual-capacity-dealer sales or dual-capacity-dealer purchases respectively. D trades not accompanied by an S or B trade are interdealer-unpaired trades. By this measure, 34.2% of B trades and 39.3% of S trades in our dataset go through agent-dealers, the remainder directly with dual-capacity-dealers. Similarly 58.6% of D trades can be duly paired and signed; the remaining 41.4% are interdealer-unpaired. Bao et al. (2016) present evidence that riskless principal trading increased following the imposition of the Volcker rule, so the agency-dealer percentages are likely even higher today. As Sirri (2014) notes and documents, agent-dealers may wait longer than a minute to pass the trade on to a principal dealer though this involves some risk. We also conducted our analyses pairing same size trades within 10 minutes. This increases the number of trades counted as agent-dealer trades about ten percent and reduces the number of dual-capacity-dealer trades accordingly. The results are almost identical to those reported below for our one-minute criterion.

Statistics on the number of trades in each classification are reported in Table 2 along with statistics on trade size (in bonds).[17] As reported in Table 2, trades that go through an agent-dealer tend to be much smaller than trades between dual-capacity-dealer and their customers. For instance, the mean (median) size of agent-dealer purchases is only 69 (10) bonds compared with 681 (100) bonds for dual-capacity-dealer purchases. For S trades, the means (medians) are 42 (15)

[16] In the few cases when the two D trades are not at exactly the same price, we drop the trade at the intermediate price. While rare, we pair multiple B or S trades with one D trade if all occur within one minute of each other and the sum of the B or S trades exactly matches the size of the D trade, e.g., two B trades for 10 bonds each and one D trade for 20 bonds. For this reason the number of agent-dealer purchase trades slightly exceeds the number of interdealer purchases and median and mean trade sizes are slightly smaller – similarly for agent-dealer sales and interdealer sales.

[17] In our version of TRACE, investment grade (speculative grade) bond trades of more than \$5 (\$1) million par value – i.e., 5000 (1000) bonds – are reported as 5MM+ (1MM+). These are 1.37% (3.96%) of trades in our sample. Since these trade sizes are truncated at 5000 (1000 bonds), the means and standard deviations in Table 2 for dual-capacity-dealer sales and purchases are understated. Means and standard deviations for the other four classifications are unbiased since there are no truncated observations in these classifications. All medians are, of course, unaffected.

bonds for agent-dealer sales and 498 (40) bonds for dual-capacity-dealer sales. In section 7 below, we find that on very small trades, total explicit transaction costs are roughly the same whether the customer deals directly with a dual-capacity-dealer or routes her trade through an agent-dealer, in which case she likely has lower internal search costs. For larger size trades, monetary transaction costs are considerably lower if the customer deals directly with a dual-capacity-dealer. Hence it is not surprising that many small trades involve an agent-dealer and that most large trades do not.

As explored by Choi and Hui (2017), liquidity may be provided by customers as well as dealers. A dual-capacity-dealer who receives a sell order but does not wish to keep the bonds in inventory may first search for a non-dealer buyer and then execute both trades simultaneously. In this case (and cases in which the dual-capacity-dealer finds a customer match for a buy order) the dual-capacity-dealer is serving in an agent-dealer role. Choi and Hui (2017) find the incidence of such trades in which customers are providing liquidity has increased since the financial crisis. Viewing same size dual-capacity-dealer S and B trades within one minute of each other as likely cases in which customers, rather than dealers, provided liquidity, we find 51,012 such pairs. These are included in Tables 2 and 3, but since they account for only 1.75% of our trade sample and 4.52% of our dual-capacity-dealer sample, and since our interest is in cases in which dual-capacity dealers function as market-makers, we exclude these trades from our dual-capacity-dealer subsequent analyses of dual-capacity dealer spread determinants.

3. Hypothesized determinants of dealer spreads

In this section, we develop hypotheses concerning the likely determinants of agent-dealer, principal-dealer, and dual-capacity-dealer spreads -- as well as the customer interface costs on dual-capacity-trades. Considered determinants fall into five groups: 1) measures of general bond market risk at the time, 2) the traded bond's liquidity, 3) measures of the bond's specific risk and information asymmetry, and 4) trade characteristics, specifically whether the bond was a likely automated electronically-executed trade, and 5) trade size. These are described in the next subsection; and the hypothesized relations for different spreads are discussed in the following subsections.

3.1. Independent variables

We expect market-making costs, but not agent costs, to be a function of the market-maker's inventory and asymmetric information risks which comprise both general corporate bond price risk at the time of the trade and the riskiness of the individual bond. To measure general corporate bond market risk at the time of the trade, we employ the MOVE and VIX indices. Merrill Lynch's MOVE index is a weighted average of the implied volatilities on 2, 5, 10, and 30-year Treasury bond options and thus is an index of expected interest rate volatility. Since corporate bonds have default as well as interest rate risk, we also employ the better known VIX index which measures implied volatility on S&P500 index options. The two indices are fairly highly correlated with $\rho = 0.769$. Since our November 2008 - December 2010 data period includes part of the financial crisis period, in some estimations we also include zero-one dummy variables for: 1) November and December 2008, 2) the first quarter of 2009, 3) second quarter of 2009 and 3) the second half of 2009. 2010 is the left-out period.

Our bond specific risk variables are the bond's rating and term-to-maturity, expecting risk and information asymmetry to be higher on low rated bonds, and price risk to be higher on longer maturity bonds. For the rating variable, we use the average of Moody's and S&P ratings from the FISD database, where AAA=1, AA+=2,.... C=22, and D=25. Thus, a higher number means a lower rating. Separately, we include a dummy variable to indicate issues where both rating agencies have withdrawn or suspended their rating. Numerous studies have found that transaction costs and spreads are general higher on lower rated and longer maturity bonds.[18] Our interest is to estimate their impact on market-making and agent-related costs separately.

As noted above, many bonds are very thinly traded; indeed, on the average day, more bonds do not trade than trade. Hence, we expect a bond's liquidity to be an important determinant of spreads. Our main measure of a bond's liquidity is the log of its average daily trade volume (summed over all dealers) over the six-month period from three months before to the trade to three months after.[19] We also anticipate that the bond's liquidity will be correlated with its rating and term-to-maturity.

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[18] Such papers include: Hong and Warga (2000), Chakravarity and Sarkar (2003), Bessembinder, Maxwell, Venkataraman (2006), EHP (2007), Harris (2015), O'Hara et al. (2016) and Choi and Huh (2017).

[19] We include trading volume over the subsequent three months so that we can include trading in recently issued bonds and so that we can explore spread determinants in the late 2008 financial crisis period. For trades in the first and last three months of our November 2008 - December 2010 period, trading volume is measured over the available data period from three to six months. By including trading volume after the trade in our trading volume measure, we are assuming that actual trading volume varies around dealers' expectations. A number of studies include similar measures and generally find that spreads decline with trading activity.

Trades on TRACE are recorded to the nearest second. Given possible rounding errors, we argue, like Harris (2015), that when the trades between an agent-dealer and customer and the accompanying trade with a principal dealer (i.e., paired S or B trades and D trades), are within a second of each other, the trades were likely automated electronically-executed trades.

Numerous studies have found that dealer spreads in general vary inversely with trade size which is what one would expect if some trading costs are fixed per trade. EHP (2007) find that the negative relation between trade size and dealer spreads is non-linear and accordingly allow a very flexible form in their estimations by including four different trade size measures. It is important to fully and accurately account for size in our data since many of our other variables are correlated with trade size. Failing to fully account for the non-linearities in the size-spread relation, could bias other coefficients. Largely following the example of EHP (2007), we regress the spreads on three trade size (measured in bonds) variables: 1) the log of trade size, 2) the reciprocal of trade size, and 3) the square root of trade size. Statistics for our continuous independent variables are presented in Table 3 on both a per-trade and per-bond basis. The latter gives more weight to less actively traded bonds. The dummy for likely automated electronically-executed trades is only measurable for agent-dealer and principal-dealer spreads for which it is equal to 1 for 59.5% of trades. We next discuss how we expect the different dealer spreads to vary with these variables.

3.2. Agent-dealer spreads

Agent-dealers essentially serve as brokers managing their customer relationships and searching among principal-dealers and other counterparties for the best price for their customers. Since they maintain no inventory, we expect agent-dealer costs and spreads to be less sensitive to the risk and asymmetric information variables than the costs and spreads of principal-dealers and dual-capacity-dealers although it is possible that agent-dealers' search costs increase if principal-dealers are reluctant to trade during high-risk periods.

On the other hand, agent-dealers' spreads should depend heavily on their search costs, which should decline with liquidity. Thus, we expect agent-dealer spreads to be inversely related to trading volume. Search costs will also depend on the agent-dealer's decision on how much effort to devote to the search process which in turn likely depends on trade size. If search costs per trade are relatively fixed, then agent-dealer spreads should be an inverse function of trade size. On the other hand, the likely incremental payoff of additional search is lower on small trades. For example, suppose the expected payoff to additional search costing \$100 is a reduction in the expected purchase price by \$0.15 per bond. This additional search is cost effective if the order is for 1000 bonds but not if it is for 10 bonds. Thus, we anticipate lower search effort on small trades which would imply lower agent-dealer spreads. Hence, we have conflicting hypotheses regarding the relation between trade size and agent-dealer spreads. Since the marginal benefit of additional search is greater, the greater the price dispersion, we also expect more search effort and larger spreads on longer maturity and lower rated bonds.

Additionally, we expect agent-dealer search costs, and hence the associated agent-dealer spreads, to be lower for automated electronically executed trades where dealers and/or their customers utilize electronic systems in which trade execution is partially or fully automated. Hence, if Harris's (2015) argument that trades paired trades within one second of each involve some form of automated electronic trading is correct, we expect a negative coefficient for the automated electronic execution proxy dummy.

3.3. Principal-dealer spreads

Since they make markets, we expect principal-dealer spreads to vary directly with their inventory risk. Thus, we anticipate higher principal-dealer spreads when corporate bond prices are more uncertain and volatile. Principal-dealers are also likely to face higher asymmetric information risk in periods of higher volatility and for more risky bonds, i.e., greater likelihood that they are trading with more informed traders. Thus, we hypothesize that principal-dealer spreads will vary directly with the MOVE and VIX indices, will be higher during the financial crisis period, and will be higher on lower-rated and longer-maturity bonds. Since inventory costs are higher when there is low turnover, we also anticipate that principal-dealer spreads will vary inversely with liquidity. Finally, to the extent that trading and order processing costs have a fixed component, we anticipate that principal-dealer spreads will vary inversely with trade size. Note that this expected trade-size-spread relationship due to fixed costs is reinforced by the economics of search discussed in section 3.2. If agent-dealers devote less search (and bargaining) to small trades, then we will tend to observe higher principal-dealer spreads on small trades, not because individual principal-dealers adjust their bid/ask prices (though they may), but because of sampling. Furthermore, it is possible that agent-dealers steer trades by less important customers, or customers with little bargaining power, to preferred (potentially higher cost) principal dealers. Note that while fixed costs and the economics of search imply different trade size – spread relations for agent-dealers, both imply that principal-dealer spreads should vary inversely with trade size. We also include the dummy variable for likely automatically electronically-executed trades. While transaction costs are likely lower, if principal dealers do not post their best prices on these exchanges, and the agent-dealer does not negotiate price improvements individually with principal-dealers, he may not find the best price, leading to higher principal-dealer spreads on automated electronically-executed trades. Finally, we include a zero-one dummy variable in the principal dealer regression to indicate those trades in which the agent-dealer spread is zero. If agent-dealers preference these trades to affiliates or favored principal-dealers (perhaps due to payment for order flow), principal-dealer spreads should arguably be higher.

3.4. Dual-capacity-dealers and their customer-interface spreads

Since (as described in section 2 above) we exclude cases where customers supplied liquidity, our remaining dual-capacity-dealers function as market-makers, and so generally face the same market-making costs as principal-dealers. Hence, we expect the hypothesized relationships discussed in section 3.3 to apply to dual-capacity-dealers as well. In other words, we expect dual-capacity-dealer spreads to vary directly with the MOVE and VIX indices, and bond term-to-maturity; we expect them to be inversely related to trading volume and trade size and to be higher on lower rated bonds. We have no way of determining if specific dual-capacity-dealer trades were automated electronically-executed trades or not, and so cannot test that variable.

Since dual-capacity-dealers trade directly with customers, they, like agent-dealers, bear costs of managing the customer relationship (which we term "customer interface costs") but do not have agent-dealer's search costs. Since dual-capacity-dealers have customer interface costs and principal-dealers do not, the difference between dual-capacity-dealer spreads and corresponding principal-dealer spreads on equivalent trades provides a measure of the dual-capacity-dealer's net customer interface costs. While managing the customer interface clearly imposes additional costs on the dual-capacity dealer, we hypothesize that dealing directly with customers provides benefits as well. Since the identity of the trader is known, the dual-capacity dealer may be able to judge whether the trader is informed or uninformed and adjust its spread accordingly. On trades routed through an agent-dealer, the principal-dealer does not have access to this information. Consequently, informed traders may choose to route their trades through agent-dealers but, anticipating this behavior, dealers functioning as principal-dealers may increase their spreads on the latter. This implies that net customer-interface-costs of dual-capacity-dealers should be lower on bonds with high information asymmetry. Thus, while we expect both principal-dealer and dual-capacity-dealer spreads to be higher on lower rated bonds, we expect net customer-interface spreads to be lower on low rated bonds. Similarly, the information asymmetry and thus the benefit from direct customer interface are likely higher on bonds which are rarely traded. Heavily traded bonds are likely have more analysts following them and private information is more likely to have already been revealed through a previous trade. Thus, while we expect both principal-dealer and dual-capacity-dealer spreads to be negatively correlated with a bond's trading volume, we expect net customer interface costs to be positively correlated. Dual-capacity-dealers also have the opportunity of using any information arising from interfacing with customers to potentially manage their inventory in that particular corporate bond more profitably.

Both the asymmetric information and inventory management benefits of direct customer interface are arguably greater in periods of high volatility, when risk and potentially the degree of information asymmetry are high. Thus, while we expect both principal-dealer and dual-capacity-dealer spreads to be positively correlated with the VIX and MOVE indices, the net customer interface costs of dual-capacity-dealers should be negatively correlated with these indices.

4. Agent-dealer and principal-dealer spreads

As noted above, the literature has taken two approaches to estimating bond dealer spreads. In the first, spreads are estimated by pairing purchase and sale prices on trades close in time. In the second, trades are not paired and spreads are estimated using a regression based on successive trades. We utilize the first procedure in this section and in section 5, and the second in section 6 and the sections that follow it.

4.1 Descriptive statistics

As noted in the introduction, the difference between the agent-dealer's sale price to the non-dealer customer and the price at which the agent-dealer simultaneously buys the bonds from a principal-dealer (in a riskless principal trade) represents the agent-dealer's compensation for searching for counterparties, for order processing, and for managing customer relationships. Likewise, the price at which an agent-dealer resells the bonds to a principal-dealer minus the price paid to the customer represents the agent-dealer's compensation for agent services on customer sale orders. The difference between a principal-dealer's sale and purchase prices represents the principal-dealer's compensation for bearing inventory and asymmetric information risks.

Average agent-dealer spreads on riskless principal sales and purchases, presented in the second and third columns respectively of Table 4, are \$0.743 and \$0.299 respectively. [20] Zero spreads are observed on 22.9% of agent-dealer sales and 51.5% of agent-dealer purchases. According to Harris (2015) and Sirri (2014), in these cases the agent-dealer is likely providing this service as part of a bundle of services to its client compensated through wrap or other fees. Consequently, Harris (2015) excludes these zero spread observations in his analysis of dealer spreads. Choi and Huh (2017) provide evidence that these zero agent-dealer spread trades are largely trades with affiliates. We choose to present results both with and without these observations. As reported in columns 4 and 5, mean agent dealer spreads are \$0.963 for sales and \$0.616 for purchases with zero-spread observation excluded.

In the final column of Table 4, we present preliminary round-trip spread statistics for principal-dealer trades with agent-dealers. For this, we pair (when possible) each principal-dealer sale trade with a principal-dealer purchase trade *on the same day*, and vice-versa, and measure the spread as the principal-dealer sale price minus the principal-dealer purchase price. When there is more than one possible purchase (sale) trade with which a sale (purchase) trade can be paired, we choose the trade closest in size and, if more than one same size trade, we choose the trade closest in time. Note that since the final column is estimated only from same day principal-dealer sales and purchases, this sub-sample tends to consist of the more actively traded bonds so the spreads in the final column should not be viewed as representative of all principal-dealer spreads. With this caveat in mind, we observe that the mean principal-dealer spread on same day trades is \$1.20, as compared average round trip agent-dealer spreads of \$1.58 (excluding zero agent-dealer spreads). Again we would caution that since the principal-dealer spread statistics in Table 4 are for actively traded bonds only, they should be regarded as illustrative only, and are not directly comparable with the agent-dealer spreads in Table 4. In the following sections, we present results on principal-dealer spreads for the universe of corporate bonds.

[20] There are several suspicious trade reports on TRACE, such as supposed trades well after the bond matured according to FISD or well before the bonds were issued. To prevent outliers possibly caused by such TRACE data errors dominating the results, all estimated spreads are winsorized at the 0.5% and the 99.5% level.

4.2. Regression results

In Table 5, regression results are presented for agent-dealer (both including[21] and excluding zero spread observations) and principal-dealer spreads. Separate coefficients were estimated for agent-dealer sales and purchases but are combined in the table (so that the coefficients measure impact on roundtrip agent-dealer spreads) for comparison with the principal-dealer spread regressions, which are necessarily roundtrip. Principal-dealer spreads are measured as the difference in the prices on principal-dealer sales to and purchases from agent-dealers which requires pairing principal-dealer sales and purchases. For each principal-dealer sale to (purchase from) an agent-dealer, we first seek a purchase (sale) on the same day. If there is no same day trade, we match the sale (purchase) with the purchase (sale) that is closest in time in either direction out to a maximum of eight weeks. 48.4% of our matches are on the same day, and 89.3% within one week. To control for interest rate changes between different trade dates, we construct a predicted price change variable based on the average percentage price change between the two trade dates for corporate bonds of the same rating and approximate maturity.[22] We control for size differences between the paired principal-dealer sale and purchase observations with separate size variables for sales and purchases which are combined in Table 5 to show the combined impact on principal-dealer spreads. Regression results with White standard errors are reported in Table 5. We estimate regressions both with and without the time period dummies which are themselves correlated with the VIX and MOVE indices.

4.2.1. Results - bond market risk

Our hypothesis that due to inventory and asymmetric information risks, principal-dealer spreads should vary directly with bond market risk is clearly confirmed. As shown in the fourth and final columns in Table 5, the coefficients of both the VIX and MOVE indices are large, positive, and significant at the .001 level. According to the coefficients in the fourth column, a one standard

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[21] We also estimated Tobit regressions the data including zero spreads. The results were qualitatively unchanged from the OLS estimations reported in Table 5.

[22] For this we calculate percentage price changes as reported on TRACE for all bonds on TRACE divided into eighteen rating/maturity bond groupings: six rating classifications: AAA & AA, A, BBB, BB, B, and below B; and three maturity groupings: 1) three to five years, 2) five to ten years, and 3) over ten years. The predicted price change for the bond is then calculated as the price on the sale date times the average percentage price change between the sale and purchase dates for bonds in the same rating/maturity group.

deviation rise in the VIX (MOVE) raises the spread about \$0.345 (\$0.242). Spreads were also higher in the financial crisis period. The results in the last column indicate that, even after controlling for the VIX and MOVE indices, for the average bond trade, principal-dealer spreads were about \$1.09 higher in late 2008 than in 2010, and declined monotonically throughout 2009.

We also hypothesized above that since they face no inventory or information risk, agent dealer spreads should be much less sensitive to the risk variables. The results in Table 5 are consistent with this. In the regressions without time dummies, the coefficients of the VIX and MOVE variables are either insignificant or negative. In the regressions with time dummies, VIX's coefficients are positive and significant but MOVE's are negative and significant and VIX's coefficients are much smaller (for instance the coefficients in column 5 imply that a one standard deviation change in the VIX is associated with only a \$0.026 change in the agent-dealer spread). Somewhat surprisingly, the coefficients of the time dummies imply that agent-dealer spreads were actually slightly lower in 2008 financial crisis period. Previous studies have found that dealer spreads are higher on low rated bonds but have not tested whether spreads vary with measures of market risk, such as the VIX and MOVE indices, and have not distinguished between agent-dealer and principal-dealer spreads.

4.2.2. Bond liquidity

Our hypothesis that, due to inventory costs, principal-dealer spreads vary inversely with how actively a bond is traded is confirmed in that principal-dealer spreads are a strong negative function of the log of average daily trade volume. A one standard deviation increase in trade volume tends to lower the principal-dealer spread about \$0.382. In the regression including all spreads, our hypothesis that agent-dealer spreads vary inversely with trading activity due to higher search costs on rarely traded bonds is also confirmed but we find that agent-dealer spreads are much less sensitive to trade volume than principal dealer spreads. The coefficient in the regression with all spreads implies that a one standard deviation increase in trade volume tends to lower agent-dealer spreads about \$0.028. Moreover, the coefficient actually becomes positive when zero spreads are excluded.[23] Several previous studies, e.g., Hong and Warga (2000), and Bessembinder, Maxwell, Venkataraman (2006), Henderschott et al. (2015), and Harris (2015) find that dealer spreads vary

[23] Zero agent-dealer spreads on dealer purchases are much more prevalent on more actively traded bonds than on rarely traded bonds. Hence, removing these from the sample raises the coefficient.

inversely with trading volume while Goldstein, Hotchkiss, and Sirri (2007) find a direct relationship. However, none heretofore have separated agent-dealer and principal-dealer spreads.

4.2.3. Rating and maturity

Previous studies have consistently found that dealer spreads in general are higher on longer maturity and lower rated bond issues but have not separated agent-dealer and principal-dealer spreads. In section 3.3, we argued that since price variability is higher on longer term and lower rated issues, principal-dealers' inventory risks are higher which should lead to higher spreads on these issues. In section 3.2 we argued that this higher price variability likely raises the expected marginal payout to additional search leading to higher search costs for agent-dealers on longer term and lower rated issues. Our Table 5 results are consistent with both hypotheses. The coefficients in the columns imply that principal-dealer spreads are about $0.23 higher on 20-year bonds than on 5-year notes and that agent-dealer spreads (including zeros) are about $0.31 higher. Our rating variable is in terms of modified ratings. In other words, the difference between A and A- rated issues is one rating unit and that between A and Baa rated issues is three. Thus, our results indicate each full (unmodified) rating drop is associated with an increase in the principal-dealer spread by about $0.14 and in the agent-dealer spread by about $0.10.

4.2.4. "Automated Electronically-Executed" trades

We argued in section 3.2 that trading costs for agent-dealers should be lower for automated electronically-executed trades (as defined in the introductory section.) Consistent with this, our dummy for likely automated electronically-executed trades is negative and significant at the .001 level implying that agent-dealer spreads are about $0.21 to $0.28 lower on automated electronically-executed trades.[24] Interestingly, this cost saving on automated electronically-executed trades is partially offset by somewhat higher ($0.10 to $0.11) principal-dealer spreads. This suggests that principal-dealers may not post their best quotes in systems where free-standing quotes can be automatically executed, or when responses to requests-for-quote are instantly generated

[24] We also tested if agent-dealer spreads were lower when the agent-dealer's trades with a customer and a principal-dealer were more than one second apart. We found slightly lower ($0.13-$0.17) agent-dealer spreads on trades two to ten seconds apart but no significant spread differences when the two trades were more than ten seconds apart. There is also evidence that the agent-dealer spread difference between electronically executed and other trades is larger on smaller trades.

algorithmically. This appears *a priori* perfectly reasonable to us. When such principal dealers provide options to the market through their free-standing quotes or quotes that have been pre-programmed in advance, these options are long-horizon options. On the other hand, when they revert with a quote in response to a specific request-for-quote in a non-automated system, the quotes that are offered have a relatively short shelf-life. In this context, it is important to note that Harris (2015) finds evidence that approximately 43% of bond market trades are not conducted at the highest available bids or lowest available asks. Our finding for principal-dealer spreads would be consistent with this.

4.2.5. Principal-dealer spreads when the agent-dealer spread is zero

As noted, in a sizable number of cases, the agent-dealer spread is zero. Choi and Huh (2017) show that these zero agent-dealer spread trades are largely trades with affiliates. We included a dummy for those cases when the agent-dealer spread was zero in the principal-dealer regression. As shown in Table 5, the evidence indicates that principal-dealer spreads are slightly but significantly higher on these trades. This could be because the principal-dealer associates dealers with an affiliate buy-side institution as trading for an institution that is potentially informed relative to generic customers unaffiliated to a major institution. However, we do not have the data to formally test this conjecture.

4.2.6. Trade size

The presence of three trade size variables in the regressions makes it difficult to determine from the coefficients in Table 5 how spreads vary with trade size. Hence, similar to EHP (2007), we use the coefficient estimates in Table 5 to estimate spreads on trade sizes ranging from 5 to 1000 bonds. The results are shown in Figure 1 assuming all non-size variables are at their sample means. EHP(2007) and others find that dealer spreads in general decline as trade size increases. We hypothesized the same for principal-dealer spreads. Consistent with this, in Figure 1, estimated principal-dealer spreads decline monotonically as trade size increases falling from \$1.54 for a trade of 5 bonds to \$0.62 for a trade of 1000 bonds.

The behavior of agent-dealer spreads as trade size increases is quite different. When zero-spreads are included, estimated spreads actually increase (and the differences are significant) until trade size reaches about 50 bonds before declining. When zero-spreads are excluded, spreads are roughly constant until trade size reaches 50 bonds. In section 3.2 we argued that the presence of fixed trading costs should cause agent-dealer spreads, like principal-dealer spreads to fall as trade size rises. However, we also hypothesized that agent-dealers might perform a less extensive search on small trades implying lower spreads on these. Since fixed costs imply a continuous decline in spreads as trade size rises, the failure of agent-dealer spreads to decline significantly until trade size reaches 50 bonds is consistent with less search on smaller trades.

5. Dual-capacity-dealer spreads

Next, we examine round-trip spreads on direct trades between dual-capacity-dealers and non-dealer customers, i.e., customer trades that do not go through an agent-dealer. As described in section 2, we drop those observations in which customers provide liquidity in order to focus on those cases where dual-capacity-dealer clearly function as market-makers. [25] As with principal-dealer spreads, we measure dual-capacity-dealer spreads as the difference between the prices of dual-capacity-dealer sales and purchases where each dual-capacity-dealer sale is matched with the dual-capacity-dealer purchase that is closest in time, and vice-versa, where the maximum time between trades is eight weeks resulting in 1,711,643 dual-capacity-dealer sale-purchase pairs.

In Table 6, we estimate basically the same dual-capacity-dealer spread regressions as were estimated in Table 5 for principal-dealer-spreads except that we cannot determine likely automated electronically-executed trades, and there is no dummy for zero agent-dealer spreads since there is no agent-dealer involvement. As noted above, non-enhanced TRACE truncates the reported trade sizes at 5000 bonds for investment grade bonds and 1000 for speculative grade. While almost no agent-dealer and principal-dealer trades were truncated, a number of dual-capacity-dealer trades are. Hence, we add zero-one dummy variables to denote truncated trade sizes on dual-capacity-dealer trades expecting negative coefficients.

As hypothesized, most of the results for dual-capacity-dealer trades are similar to those for principal-dealer trades. Again, spreads are positively related to the VIX and MOVE indices (though not for MOVE when the time dummies are included) and negatively related to trading volume although the relations are somewhat weaker than observed for principal-dealer spreads in Table 5.

[25] Regression results for the original sample of 1,780,076 are virtually identical to those reported in Table 6 and available on request. We also ran estimations removing pairs where the purchase and sale were within 15 minutes of each other. This removed another 3.24% of the sample and again the results were virtually identical to those reported in Table 6.

Similarly, spreads are higher in 2008 and early 2009 than later although the differences are not as great as in Table 5. As with principal-dealers, dual-capacity-dealer spreads are higher on lower rated and longer maturity bonds and decline as trade size increases. We will leave a more careful comparison of dual-capacity-dealer spreads with principal-dealer and agent-dealer spreads for later sections where use of a common sample allows us to test for differences.

6. All Trades: Combined Sample Evidence

Above we observed that the most common econometric approach in the literature to estimating bond transaction costs involves pairing similar sale and purchase transactions that are close in time. We followed this procedure in sections 4 and 5. The second, pioneered by Harris and Pinowar (2006) and EHP (2007), involves estimating a regression based on successive trades using regression variables to control for trade type. We next take this approach employing an extension of the EHP (2007) procedure. This has several advantages over the approach in sections 4 and 5 and most of the literature. First and most important, by estimating a model using the entire sample, instead of separate subsamples, we are able to obtain co-variances and thus statistically test for differences between agent-dealer, principal-dealer, and dual-capacity-dealer spread determinants and behaviors. Inportantly, this allows us to explore the net customer interface costs of dual-capacity-dealers. Second, we are able to reduce noise by comparing trade prices closer in time. For example, when estimating dual-capacity-dealer spreads in the previous section, we had to pair each dual-capacity-dealer sale with a dual-capacity-dealer purchase. The two trades might be far apart with many other type trades in between the two. By using price differences between successive trades regardless of type, time differences are smaller. Third, we are able to compare our spread estimates with those obtained using the EHP (2007) procedure, which does not distinguish between agent-dealer, principal-dealer, and dual-capacity-dealer trades. For sales or purchases by agent-dealers, the trade just before or just after is generally the accompanying trade between the agent-dealer and a principal-dealer. Thus, we suspect that EHP (2007)'s procedure picks up only the agent-dealer spread as the total transaction cost for those trades.

6.1. Estimation procedure

We employ a variant of the procedure developed by Harris and Piwowar (2006) and Edwards, Harris, and Piwowar (2007) (EHP). For each bond in their sample, EHP estimate:

$$r_{i,n-1,n} = \beta_{0,i}(Q_{i,n} - Q_{i,n-1}) + \sum_{j=1}^{4} \beta_{j,i}(Q_{i,n}S_{j,i,n} - Q_{i,n-1}S_{j,i,n-1}) + \sum_{k=1}^{3} \Upsilon_{k,i}\, \Delta Y_{k,i,n-1,n} + \varepsilon_{i,n-1,n} \tag{1}$$

In equation 1 (their equation 6), $r_{i,n-1,n}$ is the return on bond issue *i* between trades *n-1* and *n*. $Q_{i,n}$ is a dummy variable which =1 if trade *n* for bond *i* is a S trade (whether by an agent-dealer or a dual-capacity-dealer), = -1 if trade *n* is a B trade and =0 if a D trade (whether a sale to or purchase from an agent-dealer, or an unpaired interdealer trade). $S_{j,i,n}$ represents size measure *j* for trade *n* of bond *i*. $\Delta Y_{k,n-1,n}$ measures the change in yield index *k* from *n-1* to *n*.[26] EHP (2007) estimate equation 1 separately for each bond *i,* then calculate representative spreads using a weighted average of the coefficients.

Our approach builds on theirs but differs in several ways. First, we distinguish between agent-dealer, principal-dealer and dual-capacity-dealer spreads. Second, equation 1 assumes that all interdealer trades are equally likely to be sales or purchases while sales are more likely according to the data in Table 2. We are able to sign those interdealer trades with an agent-dealer as purchases or sales. While EHP (2007) recognize three trade types: dealer sales, dealer purchases, and interdealer trades, we recognize seven: 1) agent-dealer sales, 2) dual-capacity-dealer sales, 3) principal-dealer sales, 4) agent-dealer purchases, 5) dual-capacity-dealer purchases, 6) principal-dealer purchases, and, 7) unpaired interdealer trades. We estimate:

$$(P_{i,n} - P_{i,n-1}) = \alpha_0 + \sum_{m=1}^{6} \alpha_m (Q_{m,i,n} - Q_{m,i,n-1}) + \sum_{m=1}^{6} \sum_{j=1}^{8} \beta_{j,m}(Q_{m,n}X_{j,i,n} - Q_{m,n-1}X_{j,i,n-1}) + \Upsilon \Delta Y_{i,n-1,n} + \varepsilon_{i,n-1,n} \tag{2}$$

In equation 2, $P_{i,n}$ is the trade *n* price of bond *i* and $P_{i,n-1}$ is the price of the previous trade of the same bond.[27] The *m* subscript represents the trade type, i.e., 1) agent-dealer sales, 2) dual-capacity-dealer sales, 3) principal-dealer sales, 4) agent-dealer purchases, 5) dual-capacity-dealer purchases from non-dealers, or 6) principal-dealer purchases. $X_{j,i,n}$ represents variable *j* (e.g., VIX, bond rating, trade size, etc.) for bond *i* at the time of trade *n*. $Q_{m,i,n}$ is equal to 1 if trade *n* of bond *i* is of

[26] EHP use three indices: a general bond index, the yield difference between long and short bonds, and the difference between high and low credit risk bonds. So $\sum_{k=1}^{3} \gamma_{k,i}\, \Delta Y_{k,i,n-1,n}$ controls for changes in market rates between the two trades. As above we use a single variable based on the return on bonds of the same rating and similar maturity.

[27] EHP's dependent variable is in return form; ours in dollars. For bonds trading near par, the two measures are basically the same but differ somewhat for bonds trading at a discount or premium.

type *m* and 0 otherwise. Note that we define a dummy for each trade type except unpaired interdealer trades. Thus, $\beta_{j,m}$ estimates how variable *j* impacts the price difference between trades of type *m* and unpaired interdealer trades and $\beta_{j,m}$-$\beta_{j,p}$ estimates how variable j impacts the price difference between trades of type m and those of type p. Note, that since the sample includes all trades, agent-dealer spreads are estimated including zero-agent-dealer spreads.

EHP (2007) estimate equation 1 separately for each bond (which they are able to do since all their independent variables are time series); then average the $\beta_{j,i}$ coefficients over all bonds *I*. Since we wish to also estimate how spreads vary with cross-sectional bond characteristics, such as rating, maturity, and liquidity, we estimate equation (2) over all bonds.[28] Again we require that trades *n-1* and *n* be no more than eight weeks apart yielding a sample of 5,747,615 *n-1* and *n* trade pairs of the same bonds. The mean time between any two trades of the same bond is 0.34 days, but this is highly skewed, with 87.0% of the successive trade pairs on the same day.

6.2 Results

Estimation results are reported in Table 7. To give the results economic meaning, we report the implied impact of variable *j* on our three roundtrip spreads: agent-dealer spreads, principal-dealer spreads, and dual-capacity dealer spreads, rather than the individual coefficients, $\beta_{j,m}$. Thus, in Table 7 we report the following coefficient combinations:

1) For principal-dealer spreads: $\beta_{j,\text{principal-dealer sales}} - \beta_{j,\text{ principal-dealer purchases}}$

2) For dual-capacity-dealer spreads: $\beta_{j,\text{ dual-capacity-dealer sales}} - \beta_{j,\text{ dual-capacity-dealer purchases}}$.

3) For agent-dealer-spreads:

$(\beta_{j,\text{ agent-dealer sales}} - \beta_{j,\text{ principal-dealer sales}}) + (\beta_{j,\text{ principal-dealer purchases}} - \beta_{j,\text{ agent-dealer purchases}})$

Standard errors for the coefficient combinations corrected for heteroskedasticity and autocorrelation using the Newey-West procedure are reported in parentheses. '*' and '**' denote coefficient combinations significantly different from zero at the 5% and 1% levels respectively. The estimated impact of a one standard deviation change in the non-dummy variables is shown in brackets. Not reported in Table 7 but estimated used to construct Figure 2 are the trade size variable coefficients.

[28] Moreover, separate estimations for each bond require many observations and at least one trade of each type. Note that equation (1) has seven variables so requires at least nine observations to estimate. EHP (2007) report that this nine observation requirement eliminates approximately 20% of the bonds in their sample. For each variable *j*, we estimate separate coefficients for each trade type *m* so separate estimations of our model would require J+2 observations of each trade type m where J is the number of independent variables.

6.2.1. Market risk variables

As hypothesized, since market-making risks should increase in periods of greater price uncertainty, and paralleling our results in sections 4 and 5, principal-dealer and dual-capacity-dealer spreads are positively and strongly correlated with both the VIX and MOVE indices. Indeed, as measured by the effect of a one standard deviation change [shown in brackets in Table 7], these two variables are especially important in explaining variations in principal-dealer spreads. A one-standard deviation increase in the VIX is associated with an increase in principal-dealer spreads of \$0.240, and in dual-capacity-dealer spreads of \$0.156. A one-standard deviation increase in the MOVE index is associated with an increase in principal-dealer spreads of \$0.255, and dual-capacity-dealer spreads of \$0.043. We argued above that since they do not hold inventories, agent-dealer spreads should depend vary little, if at all with price risk. Consistent with this, as reported in column 3, agent-dealer spreads are not significantly related to either the VIX or MOVE indices.

6.2.2. Liquidity

Consistent with our previous results, and our hypothesis that both market-making and search costs are higher for illiquid bonds, Table 7 shows that principal-dealers, agent-dealers, and dual-capacity-dealers all demand more compensation for bonds with lower trading volume; all three spreads are significant negative functions of the bond's trading activity. The impact is greatest for dealers acting purely in market-making capacity, i.e., principal-dealers (for which the estimated impact of a one standard deviation change is \$0.217) and the lowest for agent-dealers.

6.2.3. Rating and maturity

Paralleling our results in sections 4 and 5, principal-dealer, agent-dealer, and dual-capacity-dealer spreads are all higher for lower rated and longer maturity bonds. As discussed above, for principal-dealers and dual-capacity-dealers this is probably due to the higher price risk on low rated and longer maturity bonds (and possibly also lower liquidity). The higher agent-dealer spreads are likely due to lower liquidity and higher search costs. As compared with five year notes, the estimates in Table 7 imply that agent-dealer roundtrip spreads on twenty year bonds are about \$0.727 higher and dual-capacity-dealer spreads about \$0.606 higher. The estimated difference in roundtrip spreads between Aa and Ba rated bonds is \$0.258 for agent-dealers, \$0.405 for principal-dealers, and \$0.339 for dual-capacity-dealers.

6.2.4. Automated Electronically-Executed Trades

As in section 4, same size agent-dealer and principal-dealer trades within one second of each other are viewed as likely automated electronically-executed trades. Results basically confirm those obtained in section 4. Consistent with lower transaction costs, agent-dealer spreads tend to be about \$0.185 lower on likely automated electronically-executed trades. Again, consistent with the results in Section 4 earlier, principal-dealer spreads tend to be about \$0.059 higher. This is consistent with agent-dealers not investing adequate resources in negotiating price improvements with principal-dealers, or with agent-dealers preferencing these trades automatically to favored principal-dealers perhaps due to payment for order flow. Either way, customers have not received the highest bid or lowest ask prices.

6.2.5. Principal-dealer spreads when the agent-dealer spread is zero

In section 4, we found evidence that principal-dealer spreads were slightly higher when the agent-dealer spread was zero consistent with agent-dealers automatically preferencing the order-flow to favored affiliates – possibly because the agent-dealer receives compensation from the principal dealer for order flow on these trades. We find stronger evidence of this in Table 7 where principal dealer spreads are found to be about \$0.167 higher on trades when the agent-dealer spread is zero. Again, customers may have not received the highest bid or lowest ask prices.

6.2.6. Trade Size

In Figure 2, we report estimated spreads for each of the three trade types by trade size based on the equation 2 regression coefficients. For these calculations, we assume that the non-size variables are at their sample means (relevant subsample means for the automated electronically-executed and zero-agent-spread variables). As in Figure 1, principal-dealer spreads decline monotonically as trade size increases. In Figure 1, when zero agent-dealer spreads were included, we again observe agent-dealer spreads rising with trade size until trade size exceeds about 50 bonds. Agent-dealer spreads on trades of 50 bonds are significantly greater than on trades of 25 bonds (or less) at the 1% level. Since the presence of fixed costs implies that spreads should decline monotonically, our interpretation of this result is that agent-dealers likely devote less search to small trades. Dual-capacity-dealer spreads decline sharply and monotonically from \$2.48 on trades of 5 bonds and \$2.25 on trades of 10 bonds to \$1.15 on trades of 100 bonds and only \$0.29 on trades of 1000 bonds. Estimated mean dual-capacity-dealer spreads are only \$0.20 on trades above 1000 bonds. Note that for smaller size trades, dual-capacity-dealer spreads substantially exceed principal-dealer spreads implying that the costs of processing trades and servicing the customer relationship (which principal-dealers avoid since these are borne by the agent-dealer) are substantial on these smaller trades. For large trades, dual-capacity-dealer spreads are actually lower than principal-dealer spreads, suggesting that for these large trades, the dual-capacity-dealer's informational advantage may more than offset the cost of servicing the customer relationship.

6.3. Customer interface benefits of dual-capacity-dealers

As argued above, since dual-capacity-dealers bear both the cost of making a market and of managing the customer relationship while principal-dealers' costs are solely market-making costs, the difference between the two spreads on equivalent bonds and trades provides a measure of the implied net customer interface costs of dual-capacity-dealers. As discussed in section 3.4, these costs are net because the direct customer interface may benefit the dual-capacity-dealer since, from her knowledge of the trader, the dual-capacity-dealer may be able to infer how informed the trader is and also may be able to use knowledge from the customer interface to manage her inventory. The principal dealer is probably denied this information since the trader employs an agent.

Referring to equation 2, we estimate the impact of factor j on the net customer interface costs of dual-capacity-dealers as:

$$(\beta_{j,\text{ dual-capacity-dealer sales}} - \beta_{j,\text{ principal-dealer sales}}) + (\beta_{j,\text{ principal-dealer purchases}} - \beta_{j,\text{ dual-capacity-dealer purchases}})$$

As discussed in Section 3.4, since information asymmetry is likely higher on low rated and rarely traded bonds, we hypothesized that net customer interface costs of dual-capacity-dealers increase with trading volume and decline as the credit rating worsens. Both hypotheses are confirmed in Panel A of Table 8. Since, the benefits of knowing the customer and how informed he is are greater when risk and information asymmetry are high, we also expect net interface costs to be negatively correlated with the VIX and MOVE indices. These hypotheses are also confirmed. A one standard deviation increase in MOVE or VIX decreases customer net interface costs (and hence arguably increases information benefits for dual-capacity-dealers from customer interface) by \$0.212 and

$0.083 respectively. Net customer interface costs are also significantly lower for bonds with greater credit risk. Finally, as shown in Figure 2 and discussed in section 6.2, Panel B of Table 8 documents that net customer interface costs are relatively large in magnitude for small trades and decrease sharply on a per-bond basis as trade size increases. This is consistent both with customer interface costs having a fixed per trade component and/or with dual-capacity-dealers deriving more informational benefits from knowing the trader's identity on larger trades.

7. Comparing transaction costs on trades with and without agent-dealers

We now explore how trading costs differ if a customer deals directly with a dual-capacity-dealer versus employing an agent-dealer. Testing cost differences is complicated by the fact that we can never observe what trading costs would have been if traders had made the other choice. Also, it needs to be kept in mind that we only observe explicit trading costs. If a trader does not employ an agent, she bears search costs which we cannot observe. With these caveats in mind, we explore how estimated monetary costs differ.

Based on our estimations of equation 2, in Panel A of Table 9, for various trade sizes we report estimated spreads on trades employing agent-dealers, consisting of both the agent-dealer and principal-dealer spreads, and on trades directly with a dual-capacity-dealer and the difference. Again, the non-size variables are set equal to their sample means for these calculations. For the smallest trades, the cost difference is miniscule and insignificant. Since employing an agent saves on the customers own search effort and expense, it appears that on these very small trades, employing an agent is likely economical. As trade size rises, transaction costs decrease rapidly for direct trades with dual-capacity-dealers, and much more slowly for trades executed through agent-dealers. Hence, for moderate and large trades, monetary costs are considerably higher if the trader employs an agent. This cost pattern is consistent with our observation in Table 2 that trades involving agent-dealers tend to be much small than those directly with dual-capacity-dealers. Nonetheless, since a large proportion of trades in the 50 to 250 bond range go through agent-dealers, it appears that monetary cost is not the sole criterion. One obvious reason is that if the trader does not employ an agent-dealer, she has to conduct her own search among dual-capacity-dealers for the best price. In addition, as discussed in section 2.4, informed traders may prefer to route trades through an agent-dealer to hide their identity from the market-maker.

Since agent-dealers bear the search costs when they are involved and the traders bear these costs when they deal directly with market makers, we expect the cost difference between trades involving an agent-dealer and trades directly with a dual-capacity dealer to be higher when search costs are high or the search is more extensive. Thus, we expect a greater cost difference for thinly traded bonds. Also, since the expected marginal benefit of additional search is higher when price dispersion is high, we expect a greater cost difference on long maturity and low rated bonds and when market uncertainty is high. Also, as observed in section 6.3, there is evidence that dual-capacity-dealers derive more benefit from knowing who the customer is when uncertainty is high In Panel B of Table 9, we report estimates of the implied impact of one unit changes in each of these variables on the cost difference. All our hypotheses are confirmed. The cost difference is significantly higher on (1) bonds with low trading volume, (2) longer maturity bonds, (3) low rated bonds, and (4) when MOVE and or VIX indices are high.

8. Reconciling the transaction cost estimates in previous studies with each other and ours

As noted in section 1, most empirical estimates of bond transaction costs have been based on either of two databases: (1) the TRACE database used here, or (2) insurance company trades from the National Association of Insurance Commissioners (NAIC). Transaction costs estimates based on the insurance data tend to be considerably lower than those based on TRACE. Sample overall roundtrip transaction cost estimates per bond using the NAIC database include: Chakravarity and Sarkar (2003): about \$0.21; Bessembinder, Maxwell, and Venkataraman (2006): \$0.18-\$0.19; Schultz (2001): about \$0.27; and Hong and Warga (2000): \$0.13 for investment grade and \$0.19 for speculative grade.[29] Based on the TRACE data, EHP (2007) estimate roundtrip transaction costs at \$1.24 for the average retail size trade and \$0.48 for the average institutional size trade; for BBB rated bonds, Goldstein, Hotchkiss, and Sirri (2007) estimate spreads from \$2.35 for trades of less than ten bonds to \$0.50 for trades of 1000 or more comparing same day trades; Zitzewitz (2011) estimates average roundtrip trading costs at \$2.52 for small trades and \$0.58 for large trades with higher costs on paired trades than on unpaired. The half-spreads of \$0.40 to \$0.42 in Bessembinder et al. (2017) for dual-capacity-dealers imply roundtrip spreads of \$0.80 to \$0.84. Harris (2015) uses an entirely different measure – the effective half-spreads – and finds that these half-spreads are

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[29] If a study estimated the spreads in percentage terms rather than dollars, we converted to dollars assuming the bond traded at par.

about 85 (52) basis points on retail (institutional) size trades implying roundtrip execution cost of about \$1.70 and \$1.04 on \$100 par value. Certainly, much of the difference between costs estimated using the TRACE and insurance company data is due to the fact that the typical insurance company trade is much larger than the typical TRACE trade but TRACE spread estimates for large trades still substantially exceed those obtained from the NAIC data. We suspect that additional reasons are: (1) that insurance companies trade primarily in investment grade bonds on which spreads are lower and (2) that insurance companies have more effective search and negotiation abilities than some other customers.

Feldhutter's (2012) transaction cost estimates using the TRACE data are much lower than those obtained by other researchers using either dataset. His imputed roundtrip transaction cost estimates range from \$0.68 for trades of 10 bonds to \$0.23 for trades of 1000 bonds. Given our findings, the reason is readily apparent. Working with data before TRACE attached the S, B, and D flags, Feldhutter was unable to identify customer buys and sales, and so estimated the trading cost as the difference between the highest trade price and the lowest where the trades were within 5 minutes of each other and for the same volume. Thus, his estimation procedure tended to pick up only the agent-dealer spread, and to exclude the principal-dealer spread, on trades involving both.

TRACE based studies which fail to account for both agent-dealer and principal-dealer trades tend to pick up one or the other but not both and thus tend to underestimate total transaction cost.[30] Consider the important paper by EHP (2007) which we have expanded in this study. EHP (2007) estimate the regression in equation 1 above using three trade types: 1) dealer sales or "S" trades (which combine our agent-dealer sales and dual-capacity-dealer sales to non-dealer customers), 2) dealer purchases or "B" trades (our agent-dealer purchases and dual-capacity-dealer purchases from non-dealer customers), and 3) interdealer or "D" trades (our trades between principal-dealers and agent-dealers, and unpaired interdealer trades). In their procedure, each trade is compared with the immediately preceding and following trade. D trades are considered as equally likely to be at the bid or ask and thus represent the expected bid-ask midpoint. Consider an

[30] It does not afflict estimates using the NAIC data since they pair customer sales and purchases and thus pick up both agent-dealer and principal-dealer spreads on trades involving both. All TRACE data spread estimates are not biased by failing to account for both spreads. Harris (2015) uses TRACE data but estimates trading costs by comparing S and B trades to quoted bid and ask prices; and Ruzza (2017) compares S&B prices to average transaction prices on the bonds that day from TRACE. By not comparing S and B trades to D trades, the transaction cost estimates in both these cases pick up both agent-dealer and principal-dealer spreads on trades involving both.

S trade. For sales by a dual-capacity-dealer, there is no problem[31] but if the S trade is a sale by an agent dealer, then either the immediately following or prior trade will normally be the agent-dealer's purchase from a principal-dealer -- in other words a D trade which is treated as equally likely to be at the bid or ask but is clearly at the principal-dealer's ask. Hence, in this case, the difference between the S and D trades is treated as representing the total spread on the transaction when in fact it only measures the agent-dealer's spread.[32] Since the trades involving an agent-dealer tend to be small, we expect this underestimation of total transaction costs to be greatest on smaller trades.

To explore how transaction cost estimates differ if costs are estimated using the TRACE data but failing to distinguish between agent-dealer, principal-dealer, and dual-capacity-dealer trades, we estimate equation 1 over the same sample with the same variables as in our earlier estimation of equation 2. Our equation 1 estimation does not duplicate exactly the procedure in EHP (2007). First, whereas they include only trade size measures as spread determinants, we include measures of bond rating, maturity, liquidity, and market uncertainty in both estimations. Second, EHP estimate equation 1 separately for each bond and average the coefficients while we estimate a single equation. Thus, comparing transaction cost estimates derived from the estimation of equation 1 with those derived from equation 2, reveals how treating all S and B trades alike (whether by agent-dealers or dual capacity-dealers) and all interdealer D trades as equally likely to be at the bid or ask (whether a principal-dealer sale or purchase), as EHP (2007) and some other studies do, impacts the transaction cost estimates but does not duplicate EHP's estimates.

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[31] Bessembinder et al. (2017) restrict their sample to dual-capacity-dealer trades only by dropping S and B trades accompanied by a D trade within one minute thus avoiding the two spread problem. Their spread estimates are comparable to ours for dual-capacity-dealer spreads.

[32] Goldstein, Hotchkiss, and Sirri (2007) estimate spreads using both paired trades and a regression and obtain considerably higher spread estimates with the latter. In their DRT spread estimate, the spread is estimated as the difference between the price at which a dealer sells bonds to a non-dealer customer and the price at which the same dealer buys the same bonds (not necessarily the same volume) from a non-dealer customer. The sample is restricted to trade pairs within 1 to 5 days and to BBB rated bonds. This sampling procedure tends to restrict the sample to trades by dual-capacity-dealers so the resulting spread estimates primarily represent dual-capacity-dealer spreads. In their "regression" procedure, Goldstein et al. (2007) utilize S and B trades so this sample adds agent-dealer trades with non-dealer customers to the DRT sample. Thus, this latter procedure picks up both agent-dealer and principal-dealer spreads on trades involving two dealers and dual-capacity-dealer spreads on trades with only one dealer so the resulting spread estimates represent total transaction costs. This would seem to explain why their regression spread estimates exceed their DRT estimates.

Resulting transaction cost estimates are shown in Figure 3. For trades of less than 100 bonds, transaction cost estimates based on equation 1 are clearly lower than those based on estimates of equation 2 and this holds both for trades that go through an agent-dealer and those directly with a dual-capacity-dealer. For trades of 100 or more bonds, the equation 1 results roughly parallel our dual-capacity-dealer spread estimates which is not surprising since the bulk of customers' larger trades are with dual-capacity-dealers.

In summary, in estimating bond transaction costs from the TRACE data, it is important to recognize that some customer trades go through agent-dealers while others are directly with market-making dealers and to use procedures which capture both agent-dealer spreads and principal-dealer spreads on the former.

9. Conclusions

The trading costs of customers in OTC security markets impound not just the costs of dealers arising from their market-making role, but also the counterparty search costs and the customer-interface related costs and benefits arising from dealers' roles as agents. We significantly extend extant empirical research on corporate bond market dealer spreads by separately estimating dealers' costs arising from their agent and market making roles, and examining their determinants in the context of the theories underpinning market-making, search costs, and the informational benefits to dealers from direct customer access.

We provide several important results. Our first major contribution is to estimate and test hypotheses on agent-dealer spreads. We find that dealers' agent-related search and net customer interface costs are sizable and comparable to dealers' market-making costs; and as hypothesized, agent-dealer spreads are not related with measures of risk (like the VIX and MOVE indices), but to measures of search costs and liquidity. In particular, agent-dealer spreads are significantly lower for automated electronically-executed trades that provide a direct proxy for lower search and order processing costs. Agent-dealer spreads also increase as bond specific trading volume decreases, and for lower rated and longer maturity bonds. Interestingly, our results on the variation of agent-related spreads with trade size are consistent with larger traders having greater bargaining power with agents, and/or agent-dealers devoting less search effort to smaller trades.

Our second contribution is to cleanly test hypotheses on market-making. Again as expected, since they bear inventory and asymmetric information risks, principal-dealer and dual-capacity-dealer spreads are significantly positively related with the VIX and the MOVE indices of volatility. And for both liquidity and risk reasons, they are also negatively related to bond specific trading volume and are higher for lower rated and longer maturity bonds. However, importantly, we find that principal-dealer spreads for automated electronically-executed trades are slightly higher, implying that the associated principal-dealers do not always post their best quotes in systems where free-standing or pre-programmed quotes can be automatically executed. This is expected since the options that principal dealers provide to the market through their free-standing or pre-programmed quotes are long-horizon options – in contrast to the short shelf-life options they provide when they revert with a quote in response to a specific request-for-quote in a non-automated framework.

Our third major contribution is to investigate the dual-capacity feature of dual-capacity-dealers. Our results are consistent with dual-capacity-dealers extracting significant information-related benefits from their direct interaction with trading customers. This benefit is greatest for large trades, where market uncertainty is high, and for issues with more information asymmetry. We also find that customers generally face significantly lower total explicit trading costs if they trade directly with a dual-capacity-dealer rather than through an agent-dealer. Since customers likely bear higher internal search costs if they do not employ an agent, future research can explore if the lower dual-capacity-dealer spreads arise because of the informational customer-interface benefits, or because customers have already incurred search costs prior to trading; or due to both factors.

Finally, we show that most earlier procedures used to estimate bond market transaction costs from TRACE data tend to significantly underestimate total transaction costs because they fail to recognize that customer transactions can often involve two component trades and spreads.

Our results have important implications for the extensive extant empirical research on dealer spreads in other OTC markets since these earlier studies have also focused solely on dealers' making-making role. This literature needs to be re-interpreted in the context of the large agent-related costs that we find, costs that meaningfully vary with economic factors in a manner similar to market-making costs in some cases, and differently in some other cases. Our results on the informational benefits to dual-capacity-dealers of directly interacting with customers – should also be applicable to other OTC markets. Finally, with traditional OTC voice-based request-for-quote markets being increasingly substituted by automated electronic execution systems, our result that principal-dealers may not post their best quotes in systems where standing quotes can be automatically executed, may again have wider applicability to other OTC markets.

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Figure 1 - Estimated Spreads of Agent-dealers and Principal-dealers by Trade Size -
based on the regression estimates in Table 5 with non-size variables at their sample means

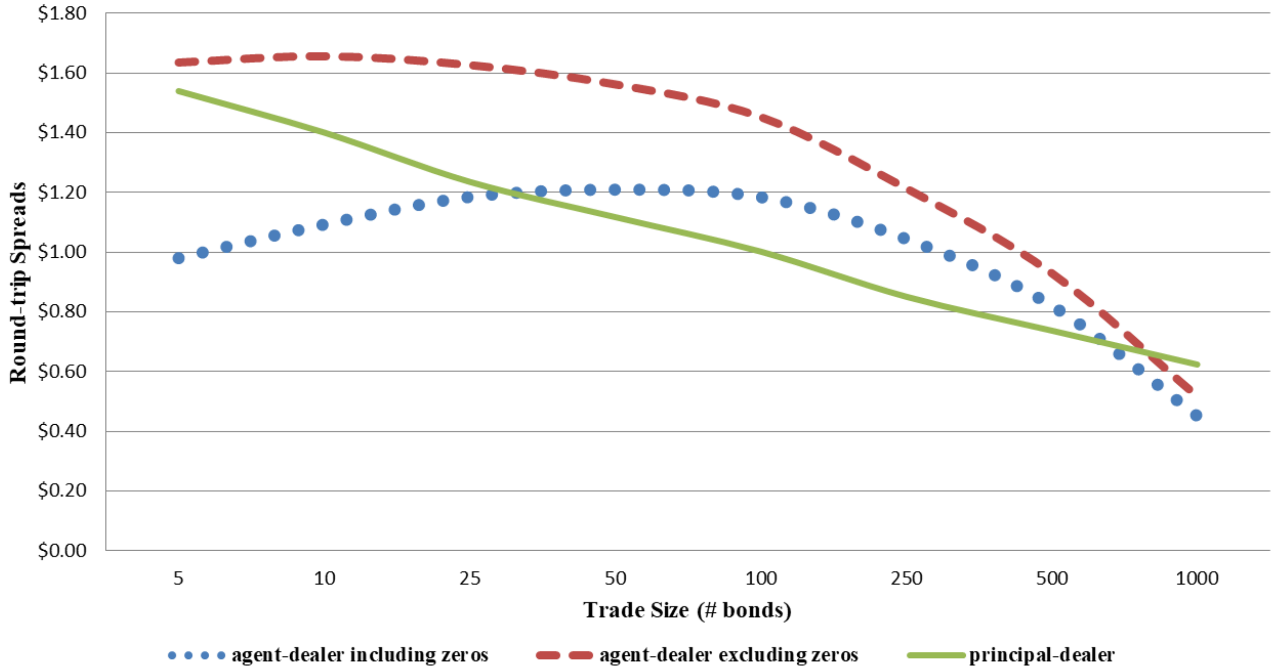

Figure 2 - Estimated spreads of dealers based on regression estimates in Table 7 with non-size variables at their sample means

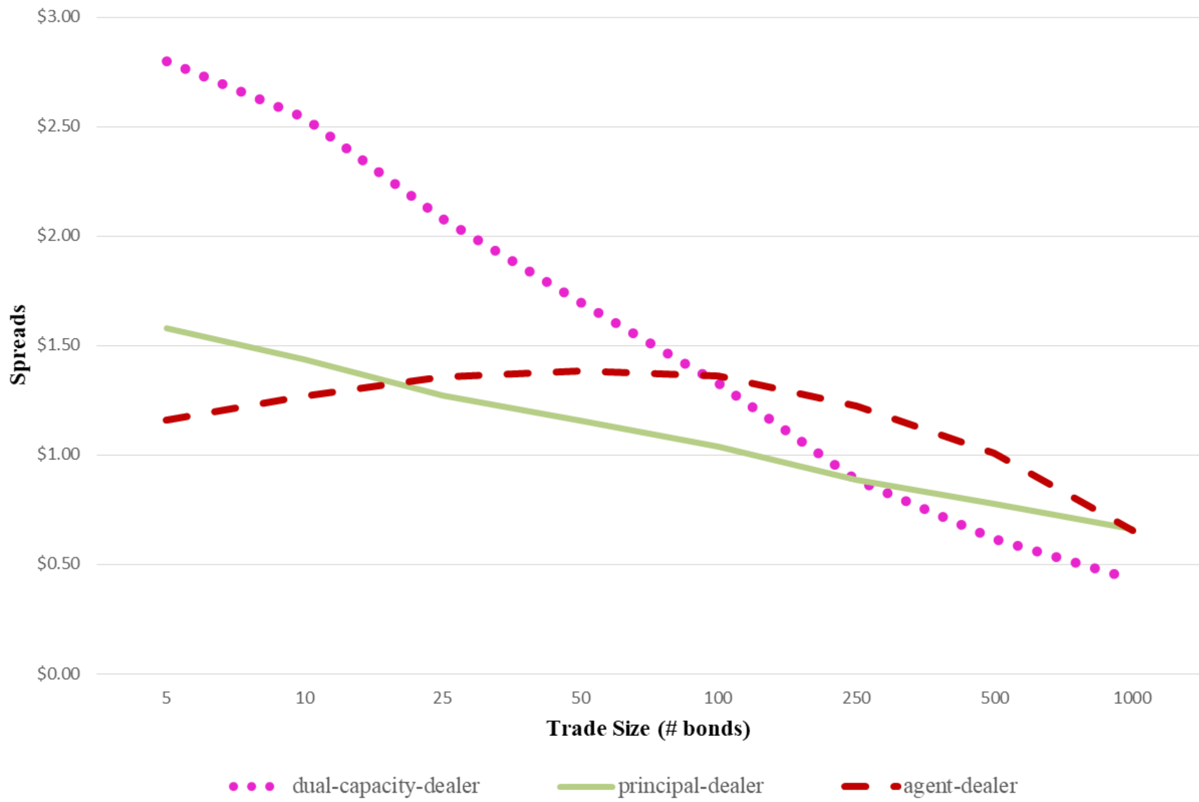

Figure 3 - Transaction cost estimates accounting and not accounting for different dealer types and trades with two dealers

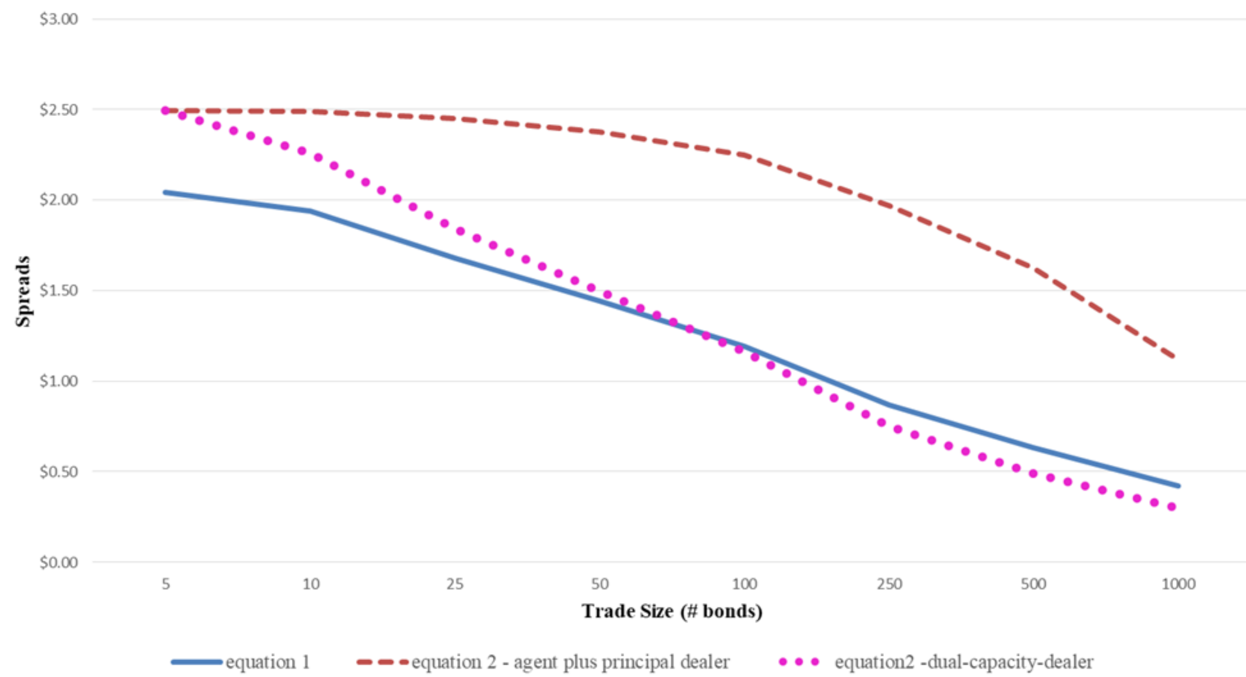

Table 1 - TRACE Data Example

Trades of Alcoa's 5.5% notes maturing Feb 1, 2017 on July 29, 2010 between 9:24 and 12:01 as reported on TRACE are shown. Columns 2-6 repeat data from TRACE. Our trade classifications are shown in the final column. We classify paired B & D trades (or S & D) trades when the trades are within one minute of each other and the quantities are identical [or the sum of the B (S) trades equals the D quantity] as "agent-dealer purchase" and "principal-dealer purchase" (or "agent-dealer sale" and "principal-dealer sale) respectively. Unpaired B and S trades are classified as "dual-capacity-dealer purchases" and "dual-capacity-dealer sales" respectively. Unpaired D trades are classified as "interdealer unpaired". The two unclassified trades in column 7 are intermediate trades or duplicates.

Trade	Time	Message sequence #	Price	Trace code	# of bonds	Classification
1	9:24:55	2798	102.309	D	10	Principal-dealer purchase
2	9:24:56	2799	101.797	B	10	Agent-dealer purchase
3	10:09:00	6027	101.028	B	25	Dual-capacity-dealer purchase
4	10:23:10	7321	102.042	D	16	Principal-dealer sale
5	10:23:10	7328	103.419	S	16	Agent-dealer sale
6	10:48:30	10062	102.5	D	3000	Interdealer unpaired
7	10:49:27	10134	102.222	D	20	Principal-dealer purchase
8	10:49:27	10137	100.722	B	20	Agent-dealer purchase
9	10:49:27	10138	102.222	D	20	
10	10:50:20	10213	102.112	D	20	Principal-dealer sale
11	10:50:20	10217	103.49	S	20	Agent-dealer sale
12	11:13:12	12935	102.749	S	25	Agent-dealer sale
13	11:13:12	12937	102.749	D	25	Principal-dealer sale
14	11:16:49	13374	103.516	S	25	Agent-dealer sale
15	11:16:52	13337	102.745	D	25	Principal-dealer sale
16	11:16:52	13340	102.745	D	25	
17	11:39:48	16259	103.901	S	10	Agent-dealer sale
18	11:39:55	16232	102.745	D	10	Principal-dealer sale
19	11:42:00	16511	101.25	B	100	Dual-capacity-dealer purchase
20	11:51:00	17524	101.75	B	5	Dual-capacity-dealer purchase
21	11:54:00	17854	103.017	S	10	Dual-capacity-dealer sale
22	12:01:00	18636	103.745	S	2	Dual-capacity-dealer sale

Table 2 - Descriptive Statistics for Paired and Unpaired Trades

Numbers of trades and statistics on trade size are reported for 5,839,480 trades of industrial bonds on TRACE between November 3, 2008 and December 31, 2010. TRACE classifies trades as purchase (B), sale (S), and interdealer (D). We classify S & D (or B & D) trades that are within one minute of each other and of identical quantityas “agent-dealer sales” (or “agent-dealer purchases”) and “principal-dealer sales” (or “principal-dealer purchases”) respectively. The remaining unpaired S, B, and D trades are termed as “dual-capacity-dealer sales”, “dual-capacity-dealer purchases”, and “interdealer unpaired” respectively. Non-enhanced TRACE truncates trade sizes at 5000 bonds (par value $5,000,000) for investment grade bonds and 1000 bonds for speculative grade bonds which affects the reported means and standard deviations, but not the medians.

			Trade size in bonds		
Trade type	# trades	% of trades	Median	Mean	Std. dev.
Dual-capacity-dealer sale	1,344,597	23.03%	40	498.1	1085.2
Agent-dealer sale	871,349	14.92%	15	42.2	179.2
Principal-dealer sale	855,873	14.66%	16	42.9	181.4
Interdealer unpaired	926,404	15.86%	90	538.0	1048.7
Principal-dealer purchase	458,737	7.86%	11	71.4	329.6
Agent-dealer purchase	472,229	8.09%	10	69.4	324.3
Dual-capacity-dealer purchase	910,291	15.59%	100	681.3	1261.0
All	5,839,480	100.00%	25	330.0	887.7

Table 3 - Continuous Independent Variables

Means, medians, and standard deviations are reported for our non-dummy independent variables for the sample of 5,832,070 trades and the sample of 3714 bonds with ratings. The latter give less actively traded bonds greater weights. The rating variable varies from AAA=1 to D=25. The mean rating of 8.47 falls between BBB+ and BBB and the mean of 9.72 between BBB and BBB-. The average daily trading volume is the average number of bonds traded per day over a six month period from three months before the trade date to three months after.

Variable	Trade sample			Bond sample		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
VIX	29.74	25.61	11.52	30.08	28.61	7.33
MOVE	119.95	111.90	34.89	120.79	117.95	21.38
Trade size (bonds)	329.54	25.00	887.43	451.99	348.67	418.82
Rating	8.47	8.50	3.80	9.72	9.16	4.09
Maturity (years)	9.42	7.02	7.65	10.26	6.77	10.02
Avg. daily trading volume (bonds)	4.62	2.51	6.05	1.72	0.64	3.74

Table 4 - Agent-dealer and Principal-dealer Spread Statistics					
Statistics are presented for agent-dealer and principal-dealer spreads on trades involving both dealer types from November 3, 2008 through December 31, 2010. Agent-dealer spreads on sales are measured as the price at which the agent-dealer sells bonds to the customer minus the price at which the agent-dealer simultaneously buys the bonds from a principal-dealer who makes a market in the bonds. Agent-dealer spreads on purchases are measured as the price at which the agent-dealer resells the bonds to a principal-dealer minus the price at which the agent-dealer simultaneously buys the bonds from the customer. Statistics are presented both including and excluding zero spreads. Principal-dealer spreads are measured as the difference between principal-dealer's sale and purchase prices on same day trades with agent-dealers. Since the principal-dealer spread sample is restricted to same day sales and purchases, it is limited to the most actively traded bonds and is thus not directly comparable to the agent-dealer spreads.					
	Agent-dealer spreads				Roundtrip principal-dealer spreads (same-day trades only)
	Including zero spreads		Excluding zero spreads		
	Sales	Purchases	Sales	Purchases	
Mean spread	\$0.743	\$0.299	\$0.963	\$0.616	\$1.195
Median	\$0.600	\$0.000	\$1.000	\$0.500	\$0.895
Standard deviation	\$0.743	\$0.517	\$0.710	\$0.597	\$1.302
Observations	871,349	472,229	672,141	229,174	530,214

Table 5- Agent-dealer and Principal-dealer Spread Regressions

Spreads on corporate bond transactions involving both an agent-dealer that organizes a riskless principal trade and a market-making principal-dealer are regressed on: bond market, bond, and trade characteristics. For principal-dealer spreads based on trades on different days, an independent variable measuring the average price change on bonds of the same rating and similar maturity is included. Coefficients of the trade characteristics (size, automated electronically-executed, and zero agent-dealer spreads) represent the estimated combined effect on both bid and ask prices. White standard errors are reported in parentheses. * and ** designate coefficients significantly different from zero at the .05 and .01 levels respectively.

	Without time dummies			With time dummies		
	All agent-dealer spreads	Agent-dealer excluding zeros	Principal-dealer spreads	All agent-dealer spreads	Agent-dealer excluding zeros	Principal-dealer Round trip
Intercept	-0.340** (.0112)	0.4007** (.0146)	1.0291** (.0201)	-0.2148** (.0126)	-0.3413** (.0166)	1.7517** (.0222)
VIX index (x100)	-0.0229 (.0127)	0.0339 (.0188)	3.0012** (.0222)	0.2278** (.0207)	0.2169** (.0284)	1.4848** (.0355)
MOVE index (x100)	-0.0125 (.0042)	-0.0285** (.0059)	0.6963** (.0068)	-0.0233** (.0057)	-0.0368** (0.0080)	0.1483** (.0088)
log of avg. daily trading volume	-0.0201** (.0008)	0.0210** (.0010)	-0.2857** (.0015)	-0.0222** (.0008)	0.0193** (.0010)	-0.2936** (.0015)
log of years to maturity	0.5227** (.0020)	0.5982** (.0025)	0.3832** (.0031)	0.5257** (.0020)	0.6004** (.0025)	0.3970** (.0031)
rating	0.0347** (.0003)	0.0273** (.0004)	0.0465** (.0005)	0.0355** (.0003)	0.0279** (.0004)	0.0492** (.0005)
dummy for withdrawn rating	0.0636** (.0126)	-0.2117** (.0140)	0.3586** (.0191)	0.0846** (.0126)	-0.1952** (0.0140)	0.4424** (.0190)
automated electronically-executed	-0.2142** (.0026)	-0.2843** (.0031)	0.0963** (.0050)	-0.2089** (.0026)	-0.2803** (.0031)	0.1147** (0.0049)
Zero agent-dealer spread			0.0183** (.0044)			0.0271** (.0043)
log of trade size (in bonds)	0.1875** (.0024)	0.0368** (.0034)	-0.1628** (.0044)	0.1877** (.0024)	0.0376** (.0034)	-0.1628** (.0044)
reciprocal of trade size	-0.3265** (.0125)	-0.3939** (.0167)	0.2608** (.0231)	-0.3189** (.0106)	-0.3894** (.0167)	0.2686** (.0230)
square root of trade size	-0.0540** (.0005)	-0.0473** (.0006)	-0.0001 (.0008)	-0.0538** (.0005)	-0.0472** (.0006)	-0.0001 (.0008)
2008 November - December				-0.1199** (.0102)	-0.0963** (.0144)	1.0904** (0.0185)
2009 first quarter				-0.0001 (.0064)	.0061 (.0091)	0.8934** (.0107)
2009 second quarter				0.0260** (.0046)	0.0214** (.0067)	0.7847** (.0074)
2009 second half				0.0679** (.0031)	0.0477** (.0040)	0.3484** (.0039)
Average price change on same rating/maturity			0.5884** (.0034)			0.5926** (.0034)
Adjusted r-square	.210	.196	.274	.212	.197	.284
Observations	1,342,823	900,741	1,087,358	1,342,823	900,741	1,087,358

Table 6 - Spread Regressions for Dual Capacity Dealer Trades

Roundtrip spreads on direct trades between a dual-capacity-dealer and a non-dealer customer are regressed on market, bond, and trade characteristics. For dual-capacity-dealer spreads based on trades on different days, a variable measuring the price change on bonds of the same rating and similar maturity is included. White standard errors are reported in parentheses. Size variable coefficients represent the estimated combined effect on both bid and ask prices. * and ** designate coefficients significantly different from zero at the .05 and .01 levels respectively.

	Without time dummies			With time dummies	
	Coefficient	Std. error		Coefficient	Std. error
Intercept	2.8689**	(.0126)		3.155**	(.0146)
VIX index (x100)	1.6600**	(.0154)		1.1274**	(.0248)
MOVE index (x100)	0.1766**	(.0049)		-0.0077	(.0064)
Log of avg. daily trading volume	-0.1240**	(.0011)		-0.1315**	(.0011)
Log of years to maturity	0.5279**	(.0018)		0.5300**	(.0018)
Average rating	0.0715**	(.0004)		0.0729**	(.0004)
Dummy for withdrawn rating	0.7726**	(.0118)		0.8205**	(.0118)
2008 November - December				0.2951**	(.0128)
2009 first quarter				0.3790**	(.0076)
2009 second quarter				0.3370**	(.0052)
2009 second half				0.1089**	(.0031)
Log of trade size (in bonds)	-0.6991**	(.0019)		-0.7058**	(.0019)
Reciprocal of trade size	-2.0332**	(.0175)		-2.1031**	(.0175)
Square root of trade size	0.0320**	(.0002)		0.0321**	(.0002)
Trade size truncated	-.4107**	(.0043)		-0.4183**	(.0043)
Average ΔP on same rating/maturity bonds	0.5591**	(.0053)		0.5593**	(.0053)
Adjusted r-square	.312			.315	
Observations	1,711,643			1,711,643	

Table 7 - Transaction Cost Relations based on Successive Trades for the Complete Sample

We estimate the equation

$$(P_{i,n} - P_{i,n-1}) = \alpha_0 + \sum_{m=1}^{6} \alpha_m (Q_{m,i,n} - Q_{m,i,n-1}) + \sum_{m=1}^{6} \sum_{j=1}^{8} \beta_{j,m} (Q_{m,n} X_{j,i,n} - Q_{m,n-1} X_{j,i,n-1}) + \Upsilon \Delta Y_{i,n-1,n} + \varepsilon_{i,n-1,n}$$

where $P_{i,n}$ is the trade n price of bond i and $P_{i,n-1}$ is the price of the previous trade of the same bond. The m subscript represents the trade type, i.e., 1) agent-dealer sales, 2) dual-capacity-dealer sales, 3) principal-dealer sales to agent-dealers, 4) agent-dealer purchases, 5) dual-capacity-dealer purchases, or 6) principal-dealer purchases from agent-dealers. $Q_{m,i,n} = 1$ if trade n of bond i is of type m and 0 otherwise. $X_{j,i,n}$ represents variable j (e.g., VIX, bond rating, trade size, etc.) for bond i at the time of trade n. For ease of interpretation, the results are reported in transaction cost form combining the effect on purchase and sale prices. Newey-West standard errors are reported in parentheses. * and ** denote parameter estimates significantly different from zero at the .05 and .01 levels respectively. As a measure of relative importance, the estimated impact on the spread of a one standard deviation change in each non-size non-dummy independent variable is shown in brackets. The estimated relation includes the usual three size variables. There are 5,747,615 observations.

	Principal-dealer spreads	Agent-dealer spreads	Dual-capacity dealer spreads
Intercepts (α_m)	0.1667** (.0264)	0.1348** (.0193)	2.5232** (.0209)
VIX index (x100)	2.0833** (.0439) [$0.240]	-0.0446 (.0229) [-$0.005]	1.3621** (.0290) [$0.156]
MOVE index (x100)	0.7305** (.0134) [$0.255]	0.0032 (.0074) [$0.001]	0.1216** (.0088) [$0.043]
log of avg. daily trading volume	-0.1620** (.0018) [-$0.217]	-0.0357** (.0016) [-$0.048]	-0.0639** (.0019) [-$0.086]
log of years to maturity	0.3090** (.0056) [$0.193]	0.5246** (.0037) [$0.328]	0.4360** (.0036) [$0.272]
average rating	0.0451** (.0009) [$0.171]	0.0282** (.0006) [$0.109]	0.0377** (.0006) [$0.143]
automated electronically-executed	0.0590** (0.0048)	-0.1846** (0.0038)	NA
Zero agent-dealer spread	0.1671** (0.0040)	NA	NA
Average ΔP for same rating/maturity bonds	0.4311** (.0046)		
Adjusted r-square	.438		

Table 8 - Determinants of the Net Cost of Direct Customer Interface for Dual-capacity-dealers

Based on the difference between spreads of dual-capacity-dealers and principal-dealers as estimated using equation 2 in Table 7, we estimate and test determinants of the benefits and costs to dual-capacity bond dealers of knowing the trader's identity. In panel A, estimates of the impact of the non-size variables on the net cost, i.e., costs minus benefits, of direct customer interface are presented. Standard errors of the estimates based on Newey-West variance-covariance matrix are shown in parentheses and estimates of the impact of a one standard deviation change in each variable in brackets. In Panel B, estimates of the net cost are reported for various trade sizes with standard error estimates in parentheses for mean values of the non-size variables . * and ** denote coefficients and estimates significantly different from zero at the 5% and 1% level respectively

Panel A – Non-size variable coefficients

VIX (x100)	MOVE (x100)	Log of trading volume	Log of years to maturity	Rating			
-0.7212** (0.0486) [-$0.083]	-0.6089** (0.0149) [-$0.212]	0.0981** (0.0025) [$0.131]	0.1270** (0.0062) [$0.079]	-0.0074** (0.0001) [-$0.028]			

Panel B - Estimated net costs for various trade sizes (bonds)

5	10	25	50	100	250	500	1000
$1.146 (0.006)	$1.025** (0.005)	$0.739** (0.005)	$0.494** (0.005)	$0.258** (0.005)	-$0.005 (0.006)	-$0.0137** (0.007)	-$0.181** (0.008)

Table 9 - Costs of Agent Assisted Trades vs Trades Directly with Market Makers

Based on the estimates of equation 2, we estimate the difference in roundtrip trading cost between trades involving an agent dealer (consisting of both the agent-dealer and principal-dealer spreads) and trades that do not (dual-capacity-dealer spreads). In Panel A we report estimated spreads for different size trades when the non-size variables are at their sample means. In Panel B we report coefficient estimates of the impact of the non-size variables on the cost difference. Standard errors based on Newey-West standard errors are shown in parentheses. * and** denote estimates significantly different from zero at the .05 and .01 levels respectively.

Panel A

Trade Size (bonds)	Agent-dealer + principal-dealer spreads	Dual-capacity-dealer spreads	Difference
5	$2.495** (0.005)	$2.494** (0.005)	$0.001 (0.006)
10	$2.492** (0.004)	$2.260** (0.004)	$0.231** (0.006)
25	$2.449** (0.004)	$1.842** (0.003)	$0.607** (0.005)
50	$2.376** (0.005)	$1.499** (0.003)	$0.877** (0.005)
100	$2.251** (0.005)	$1.161** (0.003)	$1.090** (0.006)
250	$1.969** (0.006)	$0.750** (0.003)	$1.218** (0.006)
500	$1.626** (0.008)	$0.490** (0.003)	$1.136** (0.008)
1000	$1.120** (0.012)	$0.297** (0.003)	$0.822** (0.011)

Panel B

Variable	VIX	MOVE	Log of trading volume	Log of years to maturity	Rating
	0.6766** (0.0503)	0.6057** (0.0155)	-.1338** (0.0027)	0.3976** (0.0066)	0.0356** (0.0011)